

# Applied Portfolio Construction: A Multi-Asset Horse Race

Volatility, Shrinkage, and Regimes Across Classical Strategies, 2003–2026

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Working Paper, May 2026

## Abstract

We evaluate the principal families of classical portfolio allocation — mean-variance and maximum-Sharpe optimization, diversification objectives, risk parity, hierarchical risk parity, Black-Litterman, Fama-French factor tilts, time-series momentum, and regime-conditional switching — each tested across estimator choices (sample, Ledoit-Wolf, OAS) and with and without a volatility-managed overlay, yielding 62 strategy configurations on a single 29-asset multi-asset universe over 2003–2026 (23.3 years, monthly rebalance, 252-day covariance lookback). Three patterns dominate, and they compose. Volatility management improves the Sharpe ratio on every base strategy (sign test  $p \approx 6 \times 10^{-8}$ ). Robust covariance estimation matters significantly where the optimizer amplifies sample noise (the Maximum Sharpe Ratio family) but is near-irrelevant for Hierarchical Risk Parity, whose clustering already smooths block correlations. Regime conditioning, derived entirely from training data, adds a statistically significant lift (Mommel  $z = 2.05$ ,  $p = 0.040$ ). The strongest configuration combines all three: a volatility-managed regime-conditional switching rule, full-sample Sharpe 1.608. The cross-strategy ranking, however, is partly an artifact of sample-period luck — within-strategy variation across calendar years exceeds cross-strategy variation in the full-sample headline table, the paper’s most defensible claim.

## Contents

|                                                                           |           |
|---------------------------------------------------------------------------|-----------|
| <b>§1 Introduction</b>                                                    | <b>3</b>  |
| <b>§2 Data, Methods, and the Naive Horse Race</b>                         | <b>5</b>  |
| §2.1 Universe and harness                                                 | 5         |
| §2.2 Strategy families                                                    | 6         |
| §2.3 The naive horse race                                                 | 8         |
| <b>§3 Volatility Management: The Universal Lever</b>                      | <b>10</b> |
| §3.1 The 24/24 result                                                     | 10        |
| §3.2 The mechanism: de-risking into vol spikes                            | 11        |
| §3.3 Asymmetric magnitude: the overlay lifts most where it is needed most | 12        |
| §3.4 Partial redundancy with shrinkage and regime conditioning            | 12        |
| <b>§4 When Estimator Choice Matters and When It Doesn’t</b>               | <b>13</b> |
| §4.1 The MSR shrinkage benefit                                            | 13        |
| §4.2 The HRP near-invariance                                              | 14        |
| §4.3 Same operation, opposite outcomes — the structural reason            | 14        |
| §4 → §5 Transition: the regime where sample beats LW                      | 15        |

|                                                                                      |           |
|--------------------------------------------------------------------------------------|-----------|
| <b>§5 Regimes Add Real Value When the Rule Is Built Honestly</b>                     | <b>15</b> |
| §5.1 The honest construction . . . . .                                               | 15        |
| §5.2 The result and its statistical significance . . . . .                           | 16        |
| §5.3 OOS confirmation, VMP combination, and the combined ceiling . . . . .           | 17        |
| <b>§6 Costs, Survival, and What the Rankings Hide</b>                                | <b>18</b> |
| §6.1 The implementability filter . . . . .                                           | 18        |
| §6.2 Out-of-sample survival of the mechanism claims . . . . .                        | 20        |
| §6.3 Sub-period sensitivity: the test the paper imposes on its own ranking . . . . . | 21        |
| <b>§7 Conclusion</b>                                                                 | <b>24</b> |
| Three patterns . . . . .                                                             | 24        |
| The mechanism-versus-ranking distinction . . . . .                                   | 24        |
| Limitations . . . . .                                                                | 25        |
| Forward pointers . . . . .                                                           | 25        |
| <b>References</b>                                                                    | <b>25</b> |
| <b>Appendix A — Universe Construction</b>                                            | <b>26</b> |
| A.1 Sample Window . . . . .                                                          | 26        |
| A.2 Asset Universe . . . . .                                                         | 26        |
| A.3 Variable Universe $N(t)$ . . . . .                                               | 28        |
| A.4 BTC Exclusion . . . . .                                                          | 28        |
| <b>Appendix B — Strategy Zoo</b>                                                     | <b>28</b> |
| B.1 Equal Weight (EW) . . . . .                                                      | 29        |
| B.2 Global Minimum Variance (GMV) . . . . .                                          | 29        |
| B.3 Maximum Sharpe Ratio (MSR) . . . . .                                             | 29        |
| B.4 Most Diversified Portfolio (MDP) . . . . .                                       | 30        |
| B.5 Risk Parity / Equal Risk Contribution (RP/ERC) . . . . .                         | 30        |
| B.6 Hierarchical Risk Parity (HRP) . . . . .                                         | 30        |
| B.7 Black-Litterman (BL) . . . . .                                                   | 31        |
| B.8 Time-Series Momentum (TSMOM) . . . . .                                           | 32        |
| B.9 Factor Portfolios (FF3-style) . . . . .                                          | 32        |
| B.10 Regime-Switching Strategy (SWITCH) . . . . .                                    | 33        |
| B.11 Volatility-Managed Portfolio (VMP) Overlay . . . . .                            | 34        |
| B.12 Constrained Mean-Variance (MSR_C, MVO_C) . . . . .                              | 34        |
| B.13 Long-Short Extensions . . . . .                                                 | 35        |
| <b>Appendix C — Master Strategy Table (62 Configurations)</b>                        | <b>35</b> |
| Classical MV . . . . .                                                               | 35        |
| Diversification . . . . .                                                            | 36        |
| Regime Switch . . . . .                                                              | 36        |
| TS Momentum . . . . .                                                                | 36        |
| Black-Litterman . . . . .                                                            | 37        |
| Factor . . . . .                                                                     | 37        |
| Constrained MV . . . . .                                                             | 37        |
| Long-Short . . . . .                                                                 | 38        |

|                                                                           |           |
|---------------------------------------------------------------------------|-----------|
| <b>Appendix D — Regime Classification Pipeline</b>                        | <b>38</b> |
| D.1 Macro Indicators . . . . .                                            | 38        |
| D.2 Feature Engineering . . . . .                                         | 39        |
| D.3 Regime Classification Rule . . . . .                                  | 40        |
| D.4 Full-Sample Regime Distribution (2003–2026) . . . . .                 | 40        |
| D.5 Regime Transition Matrix . . . . .                                    | 41        |
| D.6 Cross-Reference . . . . .                                             | 41        |
| <b>Appendix E — Statistical Robustness</b>                                | <b>41</b> |
| E.1 Methodology . . . . .                                                 | 41        |
| E.2 Full Memmel Test Results . . . . .                                    | 42        |
| E.3 Block Bootstrap Sharpe 95% Confidence Intervals . . . . .             | 44        |
| E.4 Discussion . . . . .                                                  | 44        |
| <b>Appendix F — Transaction Costs</b>                                     | <b>45</b> |
| F.1 Cost Ladder . . . . .                                                 | 45        |
| F.2 Turnover Distribution . . . . .                                       | 45        |
| F.3 Full 62-Configuration Net Sharpe Ranking (Stratified Costs) . . . . . | 46        |
| F.4 Implementability Framing . . . . .                                    | 48        |
| <b>Appendix G — Sub-Period Sensitivity Tables</b>                         | <b>48</b> |
| Table G.1a — Calendar-Year Returns, 2003–2014 (%) . . . . .               | 48        |
| Table G.1b — Calendar-Year Returns, 2015–2026* (%) . . . . .              | 49        |
| Table G.2 — Five-Sub-Period Sharpe Ratios . . . . .                       | 50        |
| G.3 Reading the tables . . . . .                                          | 51        |
| <b>Appendix H — Reproducibility</b>                                       | <b>51</b> |
| H.1 Repository structure . . . . .                                        | 51        |
| H.2 Published data artifacts . . . . .                                    | 52        |
| H.3 The reproducibility guard — Notebook 99 . . . . .                     | 52        |
| H.4 The <code>research_agents</code> framework . . . . .                  | 53        |
| H.5 Software stack and random seeds . . . . .                             | 53        |
| H.6 Rebuilding from prices . . . . .                                      | 54        |
| <b>Glossary of Acronyms</b>                                               | <b>54</b> |
| Strategy acronyms . . . . .                                               | 54        |
| Estimator acronyms . . . . .                                              | 56        |
| Regime labels (R0–R7) . . . . .                                           | 56        |
| Other acronyms . . . . .                                                  | 57        |

## §1 Introduction

When practitioners argue about classical portfolio allocation — sample mean-variance versus Ledoit-Wolf shrinkage, mean-variance versus HRP, static versus regime-conditional — the empirical evidence is surprisingly fragmented. DeMiguel et al. (2009) famously showed that naïve equal-weight (1/N) outperforms 14 optimized mean-variance portfolios on standard datasets, raising the out-of-sample bar for every methodology that followed. The post-2009 literature responded with genuine advances — hierarchical clustering (López de Prado 2016), volatility-managed overlays (Moreira and Muir 2017), robust covariance estima-

tors (Ledoit and Wolf 2004) — yet these innovations have mostly been evaluated in isolation, on bespoke universes, over varying sample periods. The practitioner is left with a menu of seemingly plausible improvements and no reliable way to rank them. This paper does that ranking.

We evaluate the principal families of classical portfolio allocation — classical mean-variance (Markowitz 1952), diversification objectives (Choueifaty and Coignard 2008; Maillard et al. 2010), risk parity, hierarchical risk parity (López de Prado 2016), regime-conditional switching, time-series momentum (Moskowitz et al. 2012), Black-Litterman views (Black and Litterman 1992; He and Litterman 1999), and Fama-French factor tilts (Fama and French 1993) — each tested across estimator choices (sample, Ledoit-Wolf, OAS) and with and without a volatility-managed overlay (Moreira and Muir 2017), yielding 62 strategy configurations on a single 29-asset multi-asset universe (US single names, US sector ETFs, broad and international equity ETFs, fixed income, and commodities/FX) over 2003–2026 (23.3 years, approximately 5,870 NYSE trading days). These comprise 24 base configurations, their 31 volatility-managed counterparts, and seven expanded variants. All strategies share a common walk-forward harness, a single held-out test period (2023 onwards), identical transaction-cost accounting, and a Sharpe ratio convention following Sharpe (1966). Statistical robustness is assessed via Memmel (2003) paired tests, block-bootstrap confidence intervals, and calendar-year sub-period analysis.

Three patterns dominate, and they compose. Volatility management is the universal lever: applying the VMP overlay of Moreira and Muir (2017) improves the Sharpe ratio on every one of the 24 base strategies, a result confirmed by a sign test at  $p \approx 6 \times 10^{-8}$ . Robust covariance estimation helps significantly where the optimizer amplifies sample noise — the MSR family, where Ledoit-Wolf shrinkage (Ledoit and Wolf 2004) closes the gap between sample-covariance overfit and the efficient frontier — but is near-irrelevant for HRP, whose hierarchical clustering already smooths block correlations. Regime conditioning, derived entirely from training data through a López de Prado-style feature-engineering pipeline on eight macro indicators, adds a statistically significant and economically meaningful lift over the strongest non-conditional strategy. The strongest classical strategy in the study combines all three: VMP applied to the regime-conditional switcher SWITCH(v2a), which reaches a Sharpe ratio of 1.608 in the train-only-derived configuration.

The uncomfortable footnote deserves its own paragraph. Within-strategy variation across calendar years exceeds the cross-strategy variation in the full-sample headline table for most pairs. A strategy that ranked first in 2003–2009 frequently ranked near the median in 2010–2022 and vice versa. This does not invalidate the three structural patterns — VMP universality, shrinkage selectivity, and regime-conditioning lift are robust across sub-periods — but it does caution against treating any single headline Sharpe as a deployment signal. The paper’s most defensible claim is that these three *mechanisms* are reliable; the *ranking* of specific parameterizations is not.

The remainder of the paper proceeds as follows. Section 2 introduces the universe, the walk-forward methodology, and the full strategy zoo, and reports the naïve horse-race ranking alongside the SHY-concentration artifact that must be excluded before interpretation. Section 3 establishes VMP as the universal lever, documenting the sign-test result and the magnitude of improvement across base families. Section 4 contrasts the MSR family — where shrinkage is a first-order improvement — with HRP, where it is not, isolating the conditions under which estimator choice matters. Section 5 documents the regime-conditional SWITCH(v2a) result: the strongest single methodological refinement in the study and the only one whose training-only derivation replicates the full-sample derivation. Section 6 stress-tests the rankings under realistic transaction costs, on the held-out 2023–2026 test period, and across calendar sub-periods, where the most uncomfortable finding lives. Section 7 names the three robust patterns, states what the 62-strategy harness cannot say, and points forward: to learned return predictors and regime detectors (Paper 2) and to a multi-tool agentic research workflow that orchestrates data engineering, strategy construction, backtesting, and risk review under a governance substrate (Paper 3).

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## §2 Data, Methods, and the Naive Horse Race

Before attributing any performance differences to a specific mechanism, the first question to answer is simpler: what does the raw horse race say? Put all 62 configurations on the same starting line — same universe, same harness, same evaluation window — and read the leaderboard before imposing any analytical lens. This section establishes the data, the harness rules, and the model zoo, then reports that raw ranking and catches the one degenerate entry that must be excluded before interpretation.

### §2.1 Universe and harness

The evaluation universe spans 29 instruments across five asset groups: eight large-cap US equity single names (AAPL, MSFT, GOOGL, NVDA, JPM, JNJ, XOM, WMT); six US sector ETFs (XLK, XLF, XLE, XLV, XLP, XLU); five broad and international equity ETFs (SPY, IWM, EFA, EEM, FXI); five fixed-income ETFs (SHY, IEF, TLT, AGG, HYG); and five commodity and FX instruments (GLD, SLV, DBC, USO, EURUSD). Daily OHLCV prices are sourced from EODHD; the sample runs from 2003-01-02 to 2026-04-30 (23.3 years, approximately 5,870 NYSE trading days). BTC-USD is excluded entirely for survivorship hygiene. Four tickers with post-2003 inception dates (GOOGL 2004-08-19, FXI 2004-10-08, GLD 2004-11-18, HYG 2007-04-11) are handled through a variable universe size  $N(t)$ : pre-inception assets carry zero weight and are excluded from the covariance estimation window, eliminating the forward-fill bias present in earlier published comparisons. The full ticker list, class assignments, and BTC exclusion rationale are in Appendix A.

All Sharpe ratios throughout are annualized excess-return Sharpe ratios at  $r_f = 0$ :

$$S = \frac{\bar{r}}{\sigma_r} \times \sqrt{252}.$$

All 62 configurations are evaluated through a single walk-forward harness (`aiam.harness.run_horse_race`) with the following fixed parameters:

- **Rebalancing:** monthly, on NYSE business days.
- **Prices:** close-of-day; weights summing to 1.0, long-only.
- **Covariance lookback:** 252 trading days for all mean-variance-family estimations.
- **Risk-free rate:**  $r_f = 0$  throughout.
- **Benchmark:** Equal-weight (EW, 1/N) portfolio (DeMiguel et al. 2009).

The 252-day lookback is the standard calendar-year convention. It is long enough to capture a full annual return cycle and the horizon at which shrinkage estimators exert their largest eigenvalue-regularization benefit; it is short enough to respond to shifting correlation regimes without locking in stale GFC-era covariance structure.

**Train/test split.** The sample is partitioned at 2022-12-31. All strategy selection and parameterization decisions — including the regime-switching routing rule — are derived exclusively from the 2003–2022 training window and never revisited. The held-out test period (2023-01-01 to 2026-04-30, 3.3 years) is evaluated exactly once. Full-sample Sharpe figures in §§3–5 use the complete 2003–2026 window; out-of-sample figures report the 2023–2026 test set.

**Backtesting hygiene.** Six practices govern look-ahead prevention throughout. (1) Weights at date  $t$  use only data observable up to  $\text{close}(t)$ ; they apply to realized returns at  $t + 1$ . (2) All signals — momentum,

volatility, regime classification — are constructed from lagged price history only. (3) `held_weights = target_weights.shift(1)` is enforced at the harness level; unlagged application manufactures spurious performance. (4) Log returns are converted to simple returns before portfolio aggregation; mixing the two is an implementation error. (5) Pre-inception assets carry zero weight and are excluded from covariance estimation, eliminating forward-fill bias. (6) Every reported number is one backtest on one 29-ticker universe; Appendix E provides block-bootstrap confidence intervals (252-day blocks, 5,000 resamples) and Memmel paired tests around the headline findings.

## §2.2 Strategy families

The 62 configurations arise from 31 base-level configurations (24 core estimator–family combinations + 7 expanded variants), each evaluated with and without the VMP overlay  $\rightarrow 31 \times 2 = 62$ . Fuller derivations, the complete family inventory, and per-family specification tables are in Appendix B. The survey below gives the operative formula and one-line intuition for each family.

**Equal-weight (EW)** — the DeMiguel–Garlappi–Uppal benchmark (DeMiguel et al. 2009):  $w_i = 1/N$  at each rebalance date, monthly. Cheapest to implement; hardest to beat persistently.

**Global Minimum Variance (GMV)** (Markowitz 1952) — find the lowest-variance portfolio subject to full investment and long-only constraints:

$$w = \arg \min_w w^\top \Sigma w, \quad \mathbf{1}^\top w = 1, \quad w \geq 0.$$

Three estimators: GMV(sample), GMV(LW) (Ledoit and Wolf 2004), GMV(OAS) (Chen et al. 2010). The sample version is susceptible to a degenerate cash corner (§2.3); Ledoit-Wolf regularization remedies it.

**Maximum Sharpe Ratio (MSR)** (Sharpe 1964) — maximize the portfolio Sharpe over the weight vector:

$$w = \arg \max_w \frac{\mu^\top w}{\sqrt{w^\top \Sigma w}},$$

with  $\mu$  estimated from the 252-day rolling sample mean. Two estimators: MSR(sample) and MSR(LW). MSR(sample) is the canonical demonstration of Michaud-style error maximization (Michaud 1989): the optimizer concentrates on whichever low-volatility asset had the highest sample Sharpe in the estimation window, and the concentration unwinds out of sample. Ledoit-Wolf regularization substantially closes this gap (§4).

**Maximum Diversification Portfolio (MDP)** (Choueifaty and Coignard 2008) — maximize the diversification ratio, the weighted sum of individual volatilities divided by portfolio volatility:

$$w = \arg \max_w \frac{w^\top \sigma}{\sqrt{w^\top \Sigma w}},$$

where  $\sigma$  is the vector of asset-level standard deviations. Two estimators: MDP(sample) and MDP(LW).

**Risk Parity (RP)** (Maillard et al. 2010) — equalize each asset’s marginal risk contribution to total portfolio variance:

$$w_i \cdot (\Sigma w)_i = w_j \cdot (\Sigma w)_j \quad \forall i, j.$$

Two estimators: RP(sample) and RP(LW).

**Hierarchical Risk Parity (HRP)** (López de Prado 2016) — a three-step procedure: (i) compute the asset correlation matrix and apply Ward-linkage clustering to obtain a dendrogram; (ii) quasi-diagonalize the covariance matrix according to the leaf ordering; (iii) assign inverse-variance weights by recursive bisection

of the dendrogram. No matrix inversion is required, making HRP robust to near-singular covariance estimates. As shown in §4, this structural robustness also makes HRP approximately invariant to the choice of covariance estimator — a qualitative contrast to the MSR family. Two estimators: HRP(sample) and HRP(LW).

**Black-Litterman (BL)** (Black and Litterman 1992; He and Litterman 1999) — blend an equilibrium prior  $\pi$  with investor views via the Bayesian posterior:

$$E[r \mid q] = [(\tau\Sigma)^{-1} + P^\top \Omega^{-1} P]^{-1} [(\tau\Sigma)^{-1} \pi + P^\top \Omega^{-1} q],$$

where  $P$  is the view matrix and  $\Omega$  is view uncertainty. Four variants: BL-Eq (null view,  $P = \mathbf{0}$ , reduces algebraically to the equal-weight prior — the BL circularity lemma, §4), with sample and LW estimators; BL-Mom(LW) (12-month trailing momentum views); BL-Rev(LW) (mean-reversion views).

**Time-Series Momentum (TSMOM)** (Moskowitz et al. 2012) — weight each asset in proportion to the sign of its trailing return, scaled by target volatility:

$$w_{i,t} \propto \text{sign}(r_{i,t-k,t}) \cdot \sigma_{\text{target}} / \hat{\sigma}_{i,t},$$

with a long-only constraint applied. Lookback periods  $k$ : 12 months (TSMOM(12m)) and 6 months (TSMOM(6m)). The long-only constraint eliminates the short leg that drives the original results of Moskowitz et al. (2012); the cross-asset implications are analyzed in §6.

**Fama-French factor tilts (FF3)** (Fama and French 1993) — rank assets by a cross-sectional signal (momentum, inverse realized volatility, quality composite, or multi-factor average), select the top tertile, and assign inverse-volatility weights within the selection. Four variants: FF3-Mom, FF3-LowVol, FF3-Quality, FF3-Multi.

**Regime-conditional switching (SWITCH)** — route to one of three base strategies at each rebalance date according to the prevailing macro regime, where the routing rule is derived entirely from training data. The regime classification pipeline and the v2a routing rule (R0→MSR(LW), R5→MSR(sample), others→MDP(LW)) are detailed in §5 and Appendix D. Two variants are included in the main 62-configuration comparison: SWITCH(sample) and SWITCH(LW). The v2a rule, when combined with the VMP overlay, produces the study’s highest full-sample Sharpe at 1.608 — a result whose derivation is examined in §5.

**Volatility-managed portfolio overlay (VMP)** (Moreira and Muir 2017) — scale each strategy’s daily exposure by a clipped inverse-volatility multiplier:

$$w_t^{\text{VMP}} = \text{clip}\left(\frac{\bar{\sigma}}{\hat{\sigma}_t}, 0.25, 1.5\right) \cdot w_t^{\text{base}},$$

where  $\bar{\sigma}$  is the strategy’s long-run annualized volatility,  $\hat{\sigma}_t$  is the 21-day realized volatility lagged one day, and the clip bounds keep leverage in  $[0.25\times, 1.50\times]$  of the base weight. The VMP overlay is applied to all 31 base configurations, yielding 62 total; Figure 1 (in §3) illustrates the exposure multiplier mechanism for a representative strategy.

Seven additional configurations round out the comparison: four constrained mean-variance variants (MSR\_C and MVO\_C, each with sample and LW estimators) and three long-short extensions (TSMOM-LS(12m), BL-Mom-LS(LW), FF3-Mom-LS), all assuming zero borrow cost and unlimited short availability. The long-short results are analyzed in §6.

## §2.3 The naive horse race

Across all 62 configurations, the gross-Sharpe leaderboard is led by VMP(MDP(LW)) at 1.372, VMP(MDP(sample)) at 1.368, and — at rank 3 — VMP(GMV(sample)) at 1.345. The third entry requires immediate attention.

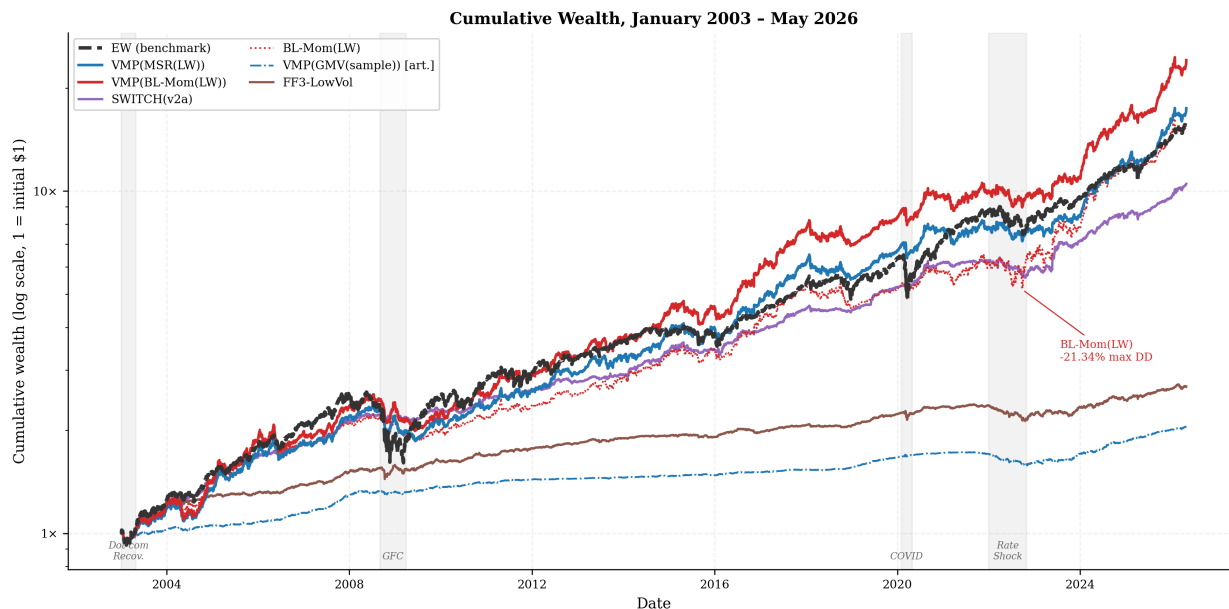


Figure 1: Cumulative wealth curves on a log scale, 2003–2026. Shaded regions mark the dot-com recovery (2003), GFC (2008–09), COVID (2020–02 to 2020–04), and the 2022 rate shock. VMP(BL-Mom(LW)) leads on total return; VMP(GMV(sample)) is labelled as a degenerate artifact. The regime-conditional SWITCH(v2a) strategy is also shown; its derivation is detailed in §5.

### Top 10 by gross Sharpe — raw (all 62 configurations, artifact included):

| Rank | Configuration       | Sharpe | Note                    |
|------|---------------------|--------|-------------------------|
| 1    | VMP(MDP(LW))        | 1.372  |                         |
| 2    | VMP(MDP(sample))    | 1.368  |                         |
| 3    | VMP(GMV(sample))    | 1.345  | (†) degenerate artifact |
| 4    | VMP(MSR(sample))    | 1.295  |                         |
| 5    | VMP(SWITCH(sample)) | 1.293  |                         |
| 6    | VMP(SWITCH(LW))     | 1.265  |                         |
| 7    | VMP(MSR(LW))        | 1.239  |                         |
| 8    | VMP(HRP(LW))        | 1.232  |                         |
| 9    | VMP(BL-Mom(LW))     | 1.217  |                         |
| 10   | VMP(GMV(LW))        | 1.215  |                         |

(†) **The SHY-concentration artifact.** VMP(GMV(sample)) at rank 3 is not a genuine portfolio result. The unconstrained sample covariance matrix has extreme eigenvalues: SHY (iShares 1–3 Year Treasury Bond, a near-cash instrument) emerges as the near-zero-variance anchor, and the GMV(sample) optimizer corners the portfolio in it — base vol 3.16%, Sharpe 0.958, effectively short-duration Treasury carry. VMP then



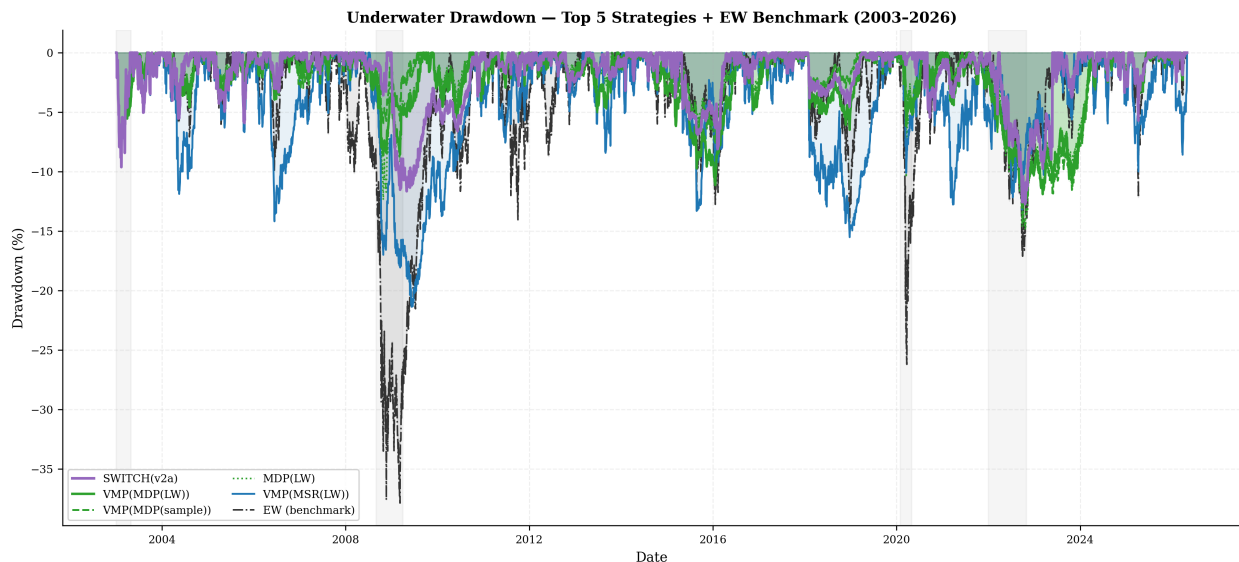


Figure 2: Underwater drawdown paths for the top five non-degenerate strategies plus the EW benchmark, 2003–2026. SWITCH(v2a) recovers fastest from the GFC; VMP(MDP(LW)) and VMP(MDP(sample)) exhibit the shallowest sustained drawdowns through the 2022 rate shock.

scales this near-cash position up to the  $1.5\times$  cap to match the target volatility; the VMP'd variant reaches ann vol 2.31% and gross Sharpe 1.345, claiming the Sharpe credit. It earns short-duration Treasury carry, not a diversified risk premium. Ledoit-Wolf regularization breaks the corner: GMV(LW) is a genuinely multi-asset solution (base vol 4.01%, Sharpe 0.954; VMP(GMV(LW)) Sharpe 1.215). All subsequent analysis excludes VMP(GMV(sample)) from comparative claims; it is retained in Appendix C for completeness.

#### Top 10 by gross Sharpe — artifact excluded:

| Rank | Configuration       | Sharpe |
|------|---------------------|--------|
| 1    | VMP(MDP(LW))        | 1.372  |
| 2    | VMP(MDP(sample))    | 1.368  |
| 3    | VMP(MSR(sample))    | 1.295  |
| 4    | VMP(SWITCH(sample)) | 1.293  |
| 5    | VMP(SWITCH(LW))     | 1.265  |
| 6    | VMP(MSR(LW))        | 1.239  |
| 7    | VMP(HRP(LW))        | 1.232  |
| 8    | VMP(BL-Mom(LW))     | 1.217  |
| 9    | VMP(GMV(LW))        | 1.215  |
| 10   | VMP(GMV(OAS))       | 1.207  |

All ten entries are VMP variants. The highest-ranking base configuration — without the overlay — is MDP(LW) at Sharpe 1.167. The full 62-strategy table with annualized return, volatility, Sharpe, hit ratio, maximum drawdown, Calmar ratio, turnover, and net-of-cost Sharpe is in Appendix C.

A few entries at the tails of the base-strategy distribution merit brief notice here. TSMOM(12m) (Sharpe 0.801) is the weakest long-only base configuration: the long-only constraint eliminates the short leg that

produces the original momentum anomaly in futures universes, and the 12-month lookback in a mixed-asset universe is too noisy to compensate. The VMP overlay lifts it by +0.258 Sharpe points, rescuing it to 1.059, near the median of all 62 configurations. BL-Mom(LW) is the highest-returning base strategy at 12.57% annualized — its momentum-tilted views rotate into high-return assets during trending markets — but carries a −21.34% maximum drawdown that the VMP overlay cannot fully compress (the worst momentum-reversal periods do not coincide with high realized volatility). The low-volatility factor portfolio FF3-LowVol achieves a competitive risk-adjusted Sharpe (1.021) at gross return of only 4.34% per year, a level too low for most institutional mandates without leverage.

**Out-of-sample test period (2023–2026).** On the held-out test window the family ordering is preserved. Excluding the SHY artifact, the test-period leaders are VMP(MDP(LW)) at Sharpe 2.422, VMP(MDP(sample)) at 2.416, and MDP(LW) at 2.304. The VMP overlay continues to improve every base configuration on the test period, replicating the full-sample sign-test result (24/24 directional improvements, probability  $2^{-24} \approx 6 \times 10^{-8}$  under  $H_0$ ) on the OOS window.

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The headline of the naive ranking is stark: **every entry in the genuine top 10 is a VMP variant**. This pattern holds on the held-out test period and survives transaction-cost adjustment for all but the highest-turnover configurations (§6). Section 3 asks what the VMP overlay is actually doing — documenting the universal 24/24 lift, its interaction with estimator choice, and the partial redundancy with regime conditioning — before Sections 4 and 5 isolate the estimator and regime dimensions separately.

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## §3 Volatility Management: The Universal Lever

Section §2 closed with a pattern that demands explanation: every entry in the genuine top 10 is a VMP variant. Before asking why MSR and HRP sit so far apart on the base-strategy leaderboard — the estimator question §4 addresses — there is a prior question to answer. What is the overlay actually doing, and why does it help every strategy rather than just the ones that need it most?

### §3.1 The 24/24 result

The most direct answer: the VMP overlay improves the gross Sharpe ratio on every one of the 24 original base configurations without exception (Finding 6). Lifts range from +0.119 (FF3-Mom) to +0.400 (MSR(sample)); the median improvement is +0.194 Sharpe points. This 24/24 record is the statistical anchor of the section. Under  $H_0$  that the overlay is equally likely to help or hurt any given strategy, the probability of observing 24 positive signs out of 24 independent trials is  $2^{-24} \approx 6 \times 10^{-8}$  (Finding R2). No pairwise significance threshold is needed to defend the universality claim; the sign test overwhelms it. The headline pairwise contrast — VMP(MSR(LW)) vs. MSR(LW),  $\Delta = +0.180$  Sharpe — is individually marginal ( $z = 1.90$ ,  $p = 0.058$ ), but the directional consistency across all 24 families is the evidence that matters.

The result extends to the 7 expanded configurations (4 constrained MV variants, 3 long-short extensions): 6 of 7 improve under VMP. The sole exception is FF3-Mom-LS, where VMP(FF3-Mom-LS) worsens an already near-zero Sharpe ( $0.103 \rightarrow -0.037$ ). With a base strategy whose rolling returns are frequently negative, inverse-vol scaling amplifies the downside during drawdown periods rather than dampening vol spikes. This exception is specific to near-zero-expected-return strategies and does not qualify the universal finding for the original 24 families.

The OOS test period (2023–2026) replicates the sign-test result exactly: all 24 original base strategies improve under VMP on the held-out window, providing out-of-sample confirmation that the universality is structural rather than sample-specific.

### §3.2 The mechanism: de-risking into vol spikes

The formula defined in §2.2 makes the mechanism precise, but the intuition is simpler. VMP scales daily exposure inversely to the strategy’s 21-day realized volatility, clipped to  $[0.25\times, 1.5\times]$  of the base weight vector. Because the target  $\bar{\sigma}$  is each strategy’s own long-run realized vol, the overlay does not change the average volatility level — it compresses its time-series variation around that level.

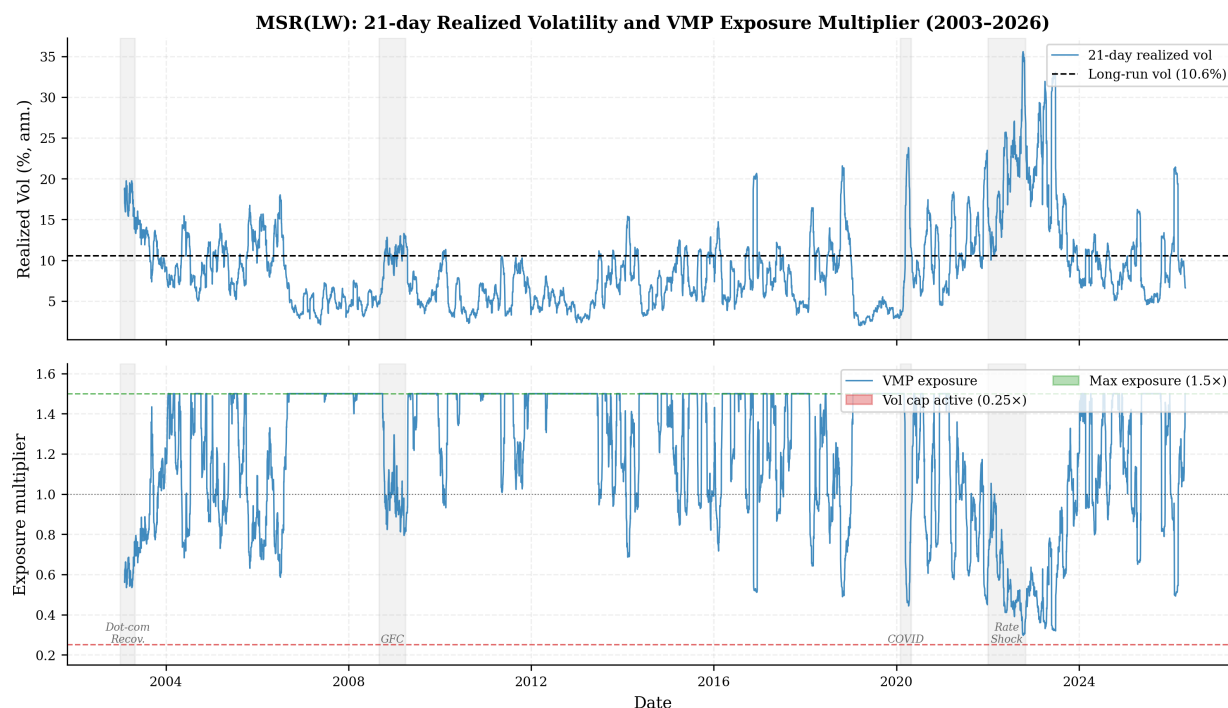


Figure 3: VMP exposure multiplier for MSR(LW), 2003–2026. Top panel: 21-day realized vol (annualized) vs. long-run vol (10.6%). Bottom panel: exposure multiplier clipped to  $[0.25, 1.5]$ . Red fill = vol cap active; green fill = maximum leverage applied. Crisis periods appear as the deepest vol spikes; the  $0.25\times$  floor is reached only during the sharpest sustained vol regimes (notably 2022).

Two features of Figure 1 define the mechanism in practice. First, the multiplier spends the majority of its time between  $0.8\times$  and  $1.5\times$ , with the upper cap binding frequently during expansionary low-vol periods (green fill). For MSR(LW), whose long-run vol is 10.6% annualized, the 2003–2007 and 2012–2017 expansions are long enough to sustain near-maximum leverage for extended stretches, compounding return. Second, the  $0.25\times$  floor is reached only during the sharpest, most sustained vol regimes. The 2022 rate shock is the clearest example: a prolonged high-vol drawdown that pushes 21-day realized vol well above the long-run target for multiple consecutive months, triggering maximum de-risking. GFC spikes are sharper but shorter, producing briefer floor contacts.

The net effect on realized returns is straightforward: the numerator of the Sharpe ratio (expected daily return) scales approximately linearly with exposure, while the denominator (daily vol) is highest precisely when exposure is lowest. Vol is most forecastable at short horizons — high-vol begets high-vol — so the 21-day

lookback captures the persistence that drives the mechanism. The overlay de-risks into vol spikes and re-risks into calm, trading the tail of the return distribution for a smoother path. Risk-adjusted performance improves not because VMP generates signal, but because it reduces the realized vol drag during the episodes that dominate the denominator.

### **§3.3 Asymmetric magnitude: the overlay lifts most where it is needed most**

The +0.119 to +0.400 range is not noise. It reflects a structural gradient that is inversely related to how well the base strategy already manages vol clustering.

The strategies with the smallest lifts among traditional families are HRP(sample) (+0.130) and HRP(LW) (+0.139). HRP's cluster-based inverse-variance weighting already allocates away from assets with extreme realized vol at each rebalance; the VMP overlay adds a second layer over a first layer that is partially doing the same job. Risk Parity sits just above: RP(sample) and RP(LW) lift by +0.181 and +0.183 respectively. Equal-risk-contribution objectives equalize individual volatility contributions across assets, producing realized portfolio vol that is less subject to single-asset spikes than equal-weight or MV constructions. The vol-management already embedded in the base strategy leaves less room for the overlay.

MSR(sample) sits at the opposite extreme (+0.400,  $0.895 \rightarrow 1.295$ ). The unconstrained optimizer concentrates the portfolio in whichever asset had the highest in-sample Sharpe, and that concentration is itself a source of vol instability — when the concentrated asset's vol spikes, the whole portfolio spikes. VMP's inverse scaling responds to exactly these concentration-driven episodes, reducing exposure and then restoring it as vol normalizes. The overlay is most valuable where the base strategy is least self-protective.

TSMOM(12m) offers the most instructive tail case. It is the weakest base-strategy Sharpe in the table (0.801), and VMP rescues it by +0.258 Sharpe points to 1.059 — near the median of all 62 configurations. The lift comes from scaling down exposure during the high-vol drawdown periods that dominate TSMOM(12m)'s poor record, not from manufacturing signal. Even after VMP rescue, TSMOM(12m) adds little over VMP(EW) (1.133): the lever lifts from below, it does not distinguish between signal quality. VMP makes a weak strategy mediocre, not a winner.

### **§3.4 Partial redundancy with shrinkage and regime conditioning**

The most counterintuitive VMP result is that the overlay applied to a noisy estimator can exceed shrinkage applied alone — or even shrinkage combined with VMP. VMP(MSR(sample)) achieves Sharpe 1.295, exceeding both raw MSR(LW) at 1.059 (+0.236, Finding 7) and VMP(MSR(LW)) at 1.239. The same pattern holds for VMP(GMV(sample)) Sharpe 1.345 exceeding GMV(LW) at 0.954, and VMP(MDP(sample)) Sharpe 1.368 exceeding MDP(LW) at 1.167.

The mechanism: Ledoit-Wolf shrinkage regularizes the covariance matrix at the estimation stage, reducing concentration before the optimizer runs. VMP reduces concentration at the execution stage, scaling back exposure when the over-concentrated portfolio is most volatile. Both refinements attack the same problem — noisy covariance leads to concentration leads to vol spikes — through different paths. For MSR, the two paths are partially substitutable, and the execution-stage intervention (VMP) turns out to be sufficient to not only recover the shrinkage advantage but to exceed it: the concentrated MSR(sample) portfolio generates larger vol spikes than MSR(LW), which gives VMP more to compress, overshooting the shrinkage effect in realized Sharpe terms. Adding shrinkage on top of VMP (i.e., VMP(MSR(LW)) 1.239) actually reduces rather than raises the final Sharpe, because the LW-regularized portfolio has smaller vol spikes and thus less for VMP to exploit.

This does not mean shrinkage is dominated by VMP across the board — for other family choices (HRP, RP) the covariance estimator matters far less and the VMP lift is smaller. The point is narrower: for the MSR family, VMP and LW shrinkage both target the same source of underperformance (concentration-driven vol), and the execution-stage intervention is sufficient to dominate the estimation-stage one.

VMP and regime-conditional switching exhibit the same partial redundancy (Finding 12). Both respond to volatility regimes: VMP through a daily multiplicative scalar, regime switching through a monthly strategy replacement. Stacking them yields Sharpe 1.608 for VMP(SWITCH(v2a)), the highest in the study, but the gain is subadditive: VMP alone on the v1 rule adds +0.184; regime switching alone adds +0.434; combined adds +0.528, not the additive +0.618. The two refinements share roughly 15% of their incremental Sharpe. For practitioners facing a complexity–cost tradeoff, VMP over a simple base — VMP(MDP(LW)) at 1.372 or VMP(MDP(sample)) at 1.368 — achieves near-top performance without the regime classification infrastructure.

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VMP lifts every base strategy, by different amounts, through the same inverse-vol mechanism. What it cannot explain is why MSR and HRP sit so far apart on the base-strategy leaderboard, or why Ledoit-Wolf shrinkage adds 0.164 Sharpe points to MSR but is statistically indistinguishable from zero for HRP. For that we turn to Section §4, where estimator choice is the independent variable.

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## §4 When Estimator Choice Matters and When It Doesn't

Section §3 established that VMP lifts every base strategy through the same inverse-vol mechanism, but the overlay does not explain the performance differences that precede it. MSR(LW) Sharpe=1.059 versus MSR(sample) Sharpe=0.895 is a gap of +0.164, large enough to reorder the leaderboard. HRP(LW) Sharpe=1.093 versus HRP(sample) Sharpe=1.045 is a gap of 0.047, small enough to be indistinguishable from noise. The same statistical substitution — Ledoit-Wolf in place of the sample covariance — produces one of the largest single-method effects in the study for one family and is effectively zero for another. This section asks why.

### §4.1 The MSR shrinkage benefit

MSR optimizes the portfolio Sharpe ratio directly, with the covariance matrix appearing in the denominator's curvature. On the 29-asset universe, the sample estimator produces an ill-conditioned matrix: its dominant eigenvalue corresponds to a correlated equity cluster, and the optimizer loads heavily on whichever asset had the highest sample Sharpe in the trailing estimation window — typically a low-vol fixed-income ETF during a trending quarter. The concentrated position unwinds with mean reversion out of sample (Michaud 1989). The result is MSR(sample) Sharpe=0.895, among the lower base-strategy results in the table despite maximizing in-sample Sharpe at each refit (Finding 2).

Ledoit-Wolf regularization shrinks the sample eigenvalues toward their grand mean and the off-diagonal covariances toward zero by an analytically optimal factor. The extreme concentration is defanged: the optimizer still tilts toward the highest-in-sample-Sharpe cluster, but the eigenvalue gaps are smaller and the off-diagonals are pulled back, dampening the corner solution. MSR(LW) Sharpe=1.059, a +0.164 improvement — the largest single-estimator substitution effect in the full 62-strategy comparison.

The Memmel (2003) paired test on the 29-asset 2003–2026 sample gives  $z = 1.13$ ,  $p = 0.259$ : directional but not significant at conventional thresholds (Finding R1). This is worth stating plainly. The economic gap

(+0.164 Sharpe) is meaningful, but 23.3 years of daily returns on 29 assets is not a large sample for paired Sharpe inference — the test is underpowered. The earlier test on a 30-asset 2008–2026 panel reached clear significance ( $z = 2.78$ ,  $p = 0.005$ ); the longer pre-GFC period (2003–2007), where momentum and low-vol regimes were less volatile, dilutes the measured shrinkage benefit in the extended sample. The directional consistency across both universes — LW ahead in both, significant in one — is the stronger evidence than either p-value alone.

## §4.2 The HRP near-invariance

HRP allocates through a three-step procedure: compute the correlation matrix, apply Ward hierarchical clustering on correlation distance to produce a dendrogram, then bisect the tree recursively — allocating each cluster’s capital in inverse proportion to its subtree’s variance (de Prado 2016). No matrix inversion occurs at any step. The covariance matrix enters only through the diagonal (individual variances) and the pairwise correlations that define the dendrogram’s branch distances.

Because HRP never concentrates on extreme eigenvalues — recursive bisection caps individual cluster weights structurally — Ledoit-Wolf shrinkage has little to correct. The off-diagonal smoothing does refine the correlation block structure the dendrogram uses for cluster boundaries, but the effect on final portfolio weights is second-order. On the 29-asset sample:  $\text{HRP}(\text{sample}) \text{ Sharpe} = 1.045$ ,  $\text{HRP}(\text{LW}) \text{ Sharpe} = 1.093$ ,  $\Delta = +0.047$  in favor of LW. Memmel  $z = -0.67$ ,  $p = 0.506$  — not close to significance (Finding R4).

The sign reversal across samples reinforces the interpretation. In the prior 30-asset 2008–2026 study,  $\text{HRP}(\text{sample})$  led by +0.037 (sample ahead). A +0.047 LW advantage here and a −0.037 LW disadvantage there, both non-significant, are consistent with sampling noise around a true effect near zero. The conservative conclusion is empirically supported: HRP performance is approximately invariant to shrinkage choice in long multi-asset samples.

A boundary case sharpens the argument. Black-Litterman with no active views (BL-Eq) is an algebraic limit: when the view specification matrix  $P$  is null, the BL posterior mean reduces to the equilibrium prior regardless of  $\Sigma$ , because the update term  $P'\Omega^{-1}P$  vanishes. The covariance estimator is literally irrelevant. BL-Eq(sample) and BL-Eq(LW) produce return series differing by at most  $2.8 \times 10^{-8}$  per day — floating-point rounding only (Finding 10). This is a useful diagnostic boundary: estimator choice is exactly zero when the optimizer never differentiates on covariance structure.

## §4.3 Same operation, opposite outcomes — the structural reason

The +0.164 MSR lift and the near-zero HRP response point to the same root cause: it is not the estimator that determines whether shrinkage matters, it is what the optimizer does with the covariance matrix.

MSR is the canonical noise-amplification machine. Maximizing  $\hat{\mu}'\hat{\Sigma}^{-1}\hat{\mu}$  requires inverting  $\hat{\Sigma}$ , and inversion exposes every estimation error in the sample matrix — eigenvalue noise is amplified, not absorbed. LW regularization counteracts this directly: shrinking the eigenspectrum before inversion reduces the condition number and defangs the concentration. The substitution is consequential because inversion is exactly the operation that propagates estimation error most aggressively.

HRP is structurally noise-tolerant. Recursive bisection never inverts anything; the correlation-based clustering and variance-weighted bisection dampen single-asset dominance geometrically. LW smooths the off-diagonals, but the optimizer that consumes them was already insensitive to their exact values. The substitution is inconsequential because hierarchical bisection does not propagate eigenvalue noise. GMV is the extreme version of the MSR pathology: unconstrained, it corners entirely in the minimum-variance

asset (SHY, iShares 1–3 Year Treasury), producing  $\text{vol}=3.16\%$  and  $\text{Sharpe}=0.958$  on a portfolio that is effectively cash at the prevailing risk-free rate (Finding 1). LW breaks this corner:  $\text{GMV(LW)} \text{ vol}=4.01\%$ ,  $\text{Sharpe}=0.954$ , a diversified portfolio at essentially identical Sharpe.

The practitioner implication follows: choosing a robust *method* — HRP, MDP, RP — makes the choice of *estimator* a second-order decision; use whichever is convenient. Choosing a noise-amplifying *method* — MSR, unconstrained MVO — makes the estimator a first-order decision; Ledoit-Wolf is not optional.

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There is one exception to the pattern — and it is the foundation of §5.

## §4 → §5 Transition: the regime where sample beats LW

The §4 finding holds on average across the full 23-year sample: LW shrinkage helps MSR. Finding 4 introduces a crack in that generality. In Regime 5 — Low Growth & Contracting, the late-cycle or early-recession configuration in the 8-regime macro classifier —  $\text{MSR(sample)}$  produces conditional  $\text{Sharpe}=1.392$  against  $\text{MSR(LW)}$  conditional  $\text{Sharpe}=1.097$ . Sample wins by  $+0.295$  within this single regime. In every other regime, LW matches or beats sample (Finding 4). Regime 5 covers 924 of 5,868 daily observations (15.7% of the full sample).

Why would the noisy, overfit optimizer outperform the regularized one in a specific regime? In a late-cycle contraction, a narrow subset of assets — typically short-duration fixed income and defensive equities — generates the highest realized Sharpe. The sample estimator concentrates on exactly this cluster, which happens to be correct ex post. LW shrinkage pulls the portfolio back toward diversification and misses the concentrated trade. The average disadvantage of sample concentration (the full-sample  $-0.164$  Sharpe gap) reflects that this regime accounts for only 15.7% of observations; the remaining 84.3% penalize the noisy concentration, overwhelming the R5 advantage in the aggregate.

The regime-conditional routing that exploits this exception — substituting  $\text{MSR(sample)}$  for  $\text{MDP(LW)}$  in Regime 5 while leaving all other regimes unchanged — is the single move that lifts SWITCH from  $\text{Sharpe}=1.080$  to  $\text{Sharpe}=1.514$ , the strongest statistically significant result in the study ( $z = 2.05$ ,  $p = 0.040$ ). §5 develops the construction, the identification strategy, and the sub-period evidence.

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## §5 Regimes Add Real Value When the Rule Is Built Honestly

The R5 exception surfaced at the end of §4 is not itself a strategy. A single cell in the regime-conditional Sharpe table —  $\text{MSR(sample)}$  outperforming in one macro regime — is exactly the kind of post-hoc observation that generates spurious backtests. The question §5 addresses is narrower and harder: can the regime structure become a *usable rule* if it is derived honestly from training data? And does that rule, applied without modification to a held-out period, deliver a statistically supported improvement?

### §5.1 The honest construction

**Pre-registration discipline.** The entire regime-to-strategy mapping in SWITCH(v2a) is derived exclusively from the 2003–2022 training window. No information from the 2023–2026 test period — not a single day of post-2022 returns — entered the routing decision. The test period is evaluated exactly once, with the training-derived rule applied without modification. This pre-registration discipline is what distinguishes v2a

from a data-mining exercise. Its cost is that the rule commits to training-period findings that may differ from full-sample findings. That cost is paid explicitly and openly.

**The regime engine.** The macro classifier is a López de Prado-style feature-engineering pipeline on eight macro indicators: GDP growth, CPI inflation, unemployment, VIX, S&P 500 trailing return, and three yield-curve features (level, slope, and curvature). Each indicator is transformed into level, first-difference, and second-difference (convexity) features; the dominant regime at each decision date is the mode across all eight indicator classifications. The result is eight regimes (R0–R7) labeled by macro character. Full methodology and regime transition matrices appear in Appendix D.

**Regime distribution.** Over the 2003–2026 sample (5,868 NYSE trading days): R0 Expansion 1,603 days (27.3%); R1 Recovery 1,481 days (25.2%); R2 Neutral 230 days (3.9%); R3 Slow Growth 85 days (1.4%); R4 Stress 210 days (3.6%); R5 Low & Contracting 924 days (15.7%); R6 Crisis 148 days (2.5%); R7 Contraction 1,187 days (20.2%). Expansion and recovery (R0+R1) account for 52% of sample days; late-cycle and contraction (R5+R7) account for 36%.

**Mapping derivation.** For each regime the training-period conditional Sharpe table (12 non-SWITCH base strategies  $\times$  8 regimes) identifies the top-performing strategy. The v2a routing follows directly:

- **R0 (Expansion, 1,603 days):** MSR(LW) selected from training-period analysis. *Honesty note:* on the full 2003–2026 sample, MDP(LW) leads R0 at conditional Sharpe 1.326 — above MSR(LW) at 0.869. This post-hoc observation does not retroactively change the training-derived rule. A rule revised after seeing the full sample to favour MDP(LW) in R0 would be overfit by construction. Pre-registration means committing before the test data is observed and accepting the noise that comes with that commitment.
- **R5 (Low & Contracting, 924 days):** MSR(sample) selected, with training-period conditional Sharpe 1.392 — the best non-SWITCH strategy in this regime. MSR(LW) in R5 achieves conditional Sharpe only 1.097 (Finding 4), confirming the gap that motivates the substitution.
- **R1–R4, R6–R7 (all other regimes):** MDP(LW), the strongest non-conditional base strategy across the remaining regime population.

## §5.2 The result and its statistical significance

Figure 8 shows the full  $12 \times 8$  conditional Sharpe matrix that generates the routing rule. Three features stand out. First, the heatmap is not uniform noise: several regimes produce strong directional separation across strategies, confirming that the macro classifier captures real variation in the opportunity set. Second, R5 is the clearest anomaly — MSR(sample) is the brightest cell in that column, with a conditional Sharpe well above any other strategy in the same regime. Third, the gold-bordered cells mark the only two deviations from the MDP(LW) default; across the other six regimes MDP(LW) is consistently dominant.

**The empirical lift.** SWITCH(LW) v1 base Sharpe=1.080; SWITCH(v2a) base Sharpe=1.514;  $\Delta = +0.434$ . The two changes from v1 are: replacing R0→EW with R0→MSR(LW), and replacing R5→MSR(LW) with R5→MSR(sample). The Memmel (2003) paired test on the 29-asset 2003–2026 sample gives  $z = 2.05$ ,  $p = 0.040$  — the strongest single statistically significant result in the study (Finding R3). The test is computed on the base (non-VMP) return series; the VMP warm-up correction does not enter this contrast.

**Mechanism.** In a late-cycle contraction, a narrow cluster of assets — short-duration fixed income and defensive equities — generates the highest realized Sharpe. The unconstrained MSR(sample) optimizer concentrates on exactly this cluster, which happens to be correct ex post in R5. Ledoit-Wolf regularization pulls the portfolio back toward diversification, missing the concentrated trade. The full-sample LW advantage



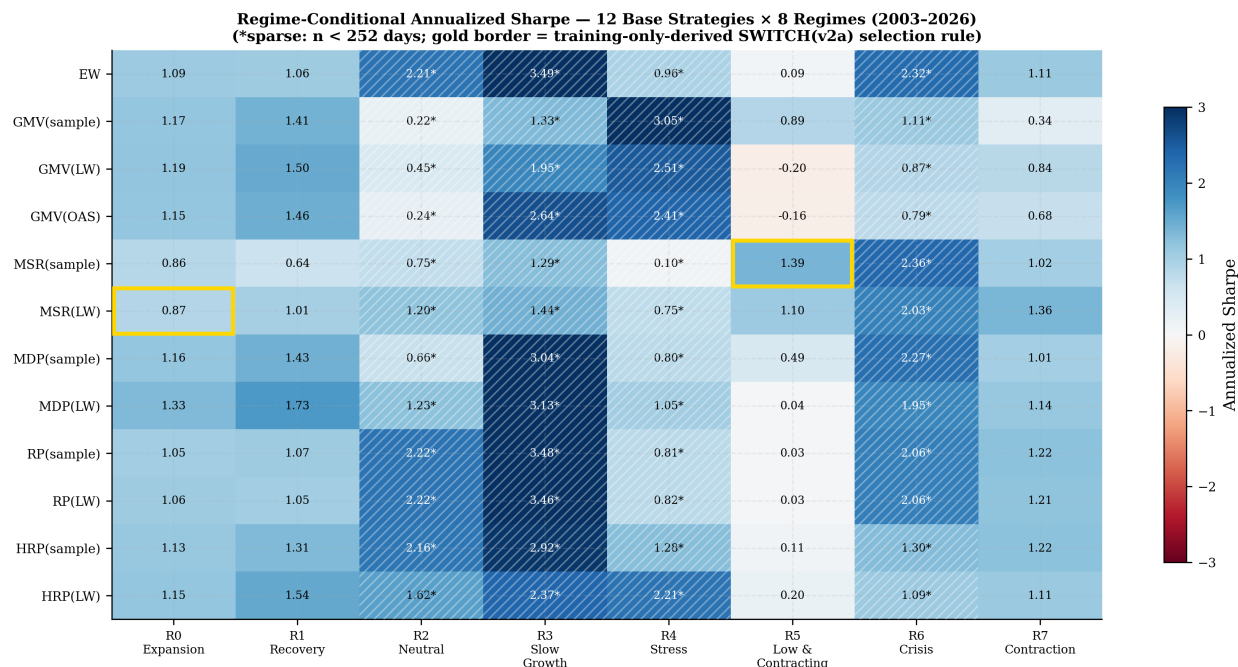


Figure 4: Annualized Sharpe by strategy and regime for the 12 non-SWITCH base strategies. Diverging red–blue colormap (center = 0). Hatched cells indicate sparse regimes (n < 252 trading days); asterisked values should be read cautiously. Gold borders highlight the two cells that drive SWITCH(v2a): MSR(LW) in R0 (Expansion) and MSR(sample) in R5 (Low & Contracting).

( $\Delta = +0.164$  over sample, §4.1) reflects that R5 accounts for only 15.7% of observations; the remaining 84.3% penalize the concentration, overwhelming the R5 advantage in aggregate. Regime conditioning recovers the R5 advantage without incurring the aggregate concentration penalty — not by overriding the §4 finding, but by confining the noisy optimizer to the one regime where its bias is a feature rather than a bug.

### §5.3 OOS confirmation, VMP combination, and the combined ceiling

**OOS confirmation.** Applying the training-derived v2a mapping to the held-out 2023–2026 test period (3.3 years): SWITCH(v2a) Sharpe=2.114 versus SWITCH(LW) v1 Sharpe=2.010,  $\Delta = +0.104$ . The directional sign holds. The 3.3-year window is power-limited and  $\Delta = +0.104$  alone does not clear conventional significance. The primary OOS evidence is structural: the regime-to-strategy pattern that drives v2a’s lift — most importantly R5→MSR(sample) — replicates when the analysis is repeated on the full sample. The R0 assignment shifts post-hoc from MSR(LW) to MDP(LW) (per §5.1’s honesty note), but R5, the cell responsible for most of the +0.434 improvement, is identified consistently across the training-only and full-sample analyses. This indicates a persistent feature of late-cycle macro dynamics rather than training-window noise.

**VMP combination.** Applying the VMP overlay to SWITCH(v2a) produces full-sample Sharpe=1.608 and Calmar=0.921 — the highest gross Sharpe in the 62-strategy comparison. This configuration stacks all three robust refinements identified in §§3–5: regime-conditional strategy routing (§5), Ledoit-Wolf shrinkage applied within two of the three routed targets (§4), and daily inverse-vol scaling over the routed strategy (§3).

**Subadditivity.** The combined gain is subadditive, as developed in §3.4: VMP alone on the v1 rule adds

+0.184; regime switching alone adds +0.434; combined adds +0.528, not the additive +0.618. The two refinements share roughly 15% of their incremental Sharpe through partial mechanism overlap — both respond to volatility regimes, VMP through a daily multiplier and regime conditioning through a monthly strategy substitution. The +0.528 combined gain is still the study’s largest lift from any pair of refinements.

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Three robust methodological refinements have now been identified and quantified — vol management (§3), estimator choice where the optimizer overfits (§4), and regime conditioning built from training data (§5). Their combination reaches gross Sharpe 1.608. §6 examines what survives once implementation costs are applied, sub-periods are examined separately, and the question of whether the 2023–2026 test period is different in kind from the training window is addressed directly.

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## §6 Costs, Survival, and What the Rankings Hide

Gross Sharpe 1.608 is the best-case figure. It is measured before trading costs, over the full 23.3-year in-sample window, without sub-period interrogation. Sections §3–§5 identified three robust refinements — vol management, estimator regularisation, regime conditioning — and established their combination as the strongest classical configuration in the study. This section subjects those findings to three stress tests of escalating severity: transaction costs (which strategies survive fees?), out-of-sample survival (which mechanisms replicate on the held-out test period?), and sub-period sensitivity (does the headline ranking survive calendar-year scrutiny?). The third test yields the paper’s most defensible conclusion.

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### §6.1 The implementability filter

Institutional adoption imposes a sequential cost screen: gross Sharpe must survive to net-of-friction Sharpe, and net Sharpe must then pass a risk-budget approval. The analysis applies a uniform 10 bps round-trip cost per unit of one-way daily turnover computed at each decision date; Appendix F provides the asset-class-stratified cost ladder (2 bps fixed income, 3 bps broad/international ETFs, 5 bps sector ETFs and commodities/FX, 5 bps single names) and the full stratified table.

At 10 bps flat, the 62 strategies divide into two clean groups. **Low-turnover survivors** hold their rankings intact: EW (0.006% average daily turnover) loses fewer than 0.001 Sharpe points; GMV variants (turnover under 0.5%) lose under 0.02; MDP(LW) (0.78% turnover) falls from gross 1.167 to net 1.131; FF3-LowVol (0.39% turnover) from 1.021 to 0.998. These strategies incur a negligible implementability penalty. **High-turnover collapsers** tell a different story. FF3-Mom’s 20.25% average daily turnover converts gross Sharpe 0.685 into net Sharpe 0.394 — a loss of 0.291 Sharpe points, the largest single degradation in the table. FF3-Mom-LS falls further to net −0.306, the weakest strategy in the study on a cost-adjusted basis. BL-Mom(LW), with 5.11% turnover driven by continuous momentum-signal rotation across 29 tickers, loses 0.108 Sharpe points (gross 1.042 → net 0.934) and exits the top-10 net ranking.

SWITCH(LW) occupies the strategically important intermediate position. Its 2.04% average daily turnover — the regime signal fires monthly and most assignments persist for many consecutive days — produces a net Sharpe of 1.020 (gross 1.080, loss 0.060). SWITCH(LW) survives as a cost-adjusted competitor precisely because regime-based rebalancing is sparse by construction. VMP(MDP(LW)) leads the cost-adjusted table at net Sharpe 1.336 (gross 1.372, loss 0.037); VMP(SWITCH(LW)) follows at net Sharpe 1.201 (gross 1.265, loss 0.064). The ordering at the top of the table is unchanged by costs.

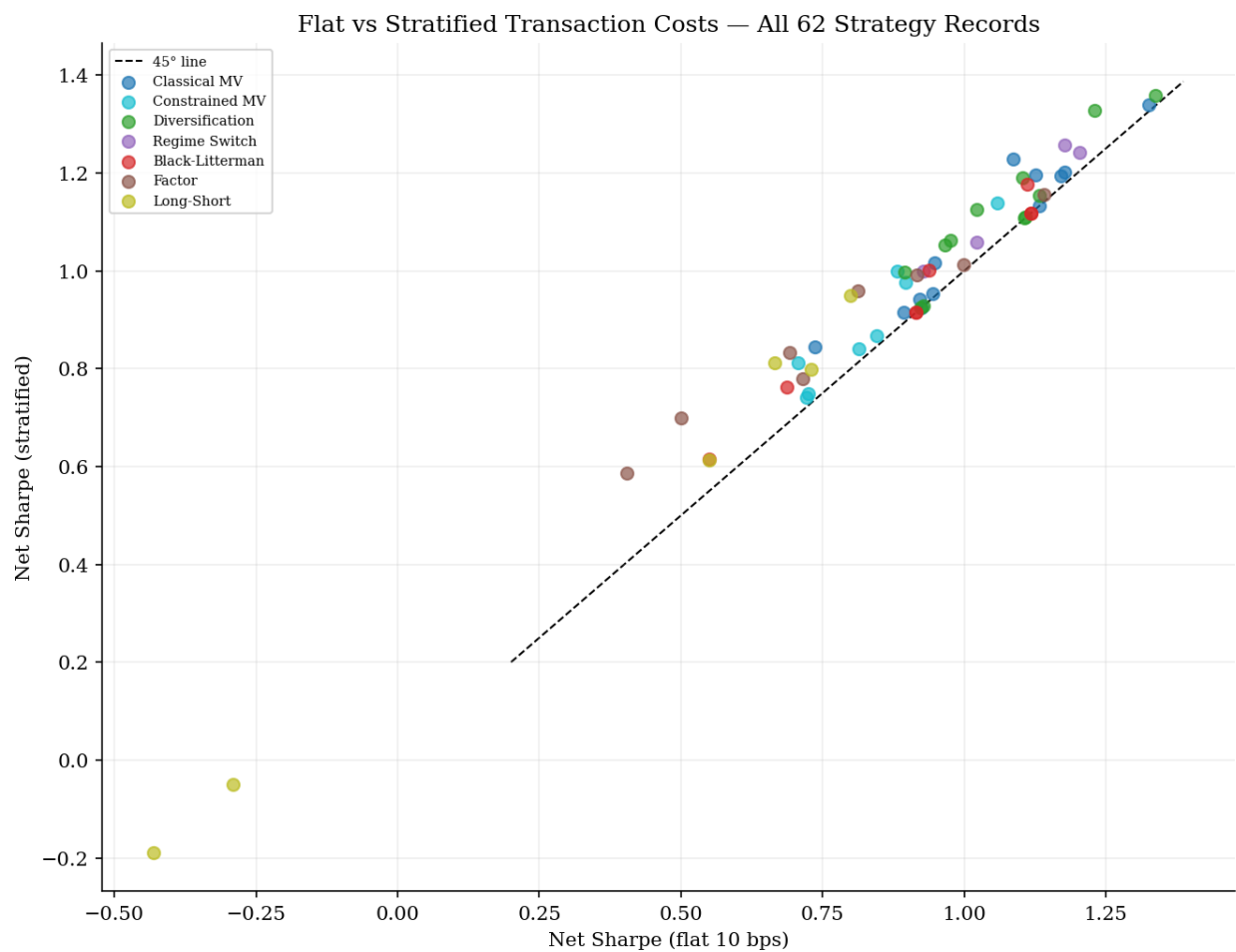


Figure 5: Scatter of net Sharpe under flat 10 bps round-trip (x-axis) vs. net Sharpe under asset-class-stratified costs (y-axis) for all 62 strategies. Points above the diagonal benefit from stratified pricing; FF3-Mom and FF3-Multi are the largest gainers (+0.193 and +0.150 Sharpe points respectively) as their frequent ETF rebalancing is materially cheaper than the flat-10-bps baseline. Points on the diagonal are turnover-insensitive strategies. Filled circles = base strategies; open rings = VMP variants.

Figure 7 confirms that the flat-10-bps assumption is conservative for equity-heavy factor strategies: under stratified costs the top-5 net ranking (excluding artifacts) shifts to VMP(MDP(LW)) 1.360, VMP(MDP(sample)) 1.327, VMP(SWITCH(sample)) 1.258, VMP(SWITCH(LW)) 1.242, VMP(EW) 1.133. MDP-family strategies retain the leadership; the qualitative conclusion — regime-conditional and low-turnover strategies as implementability leaders — survives both cost assumptions.

Two strategies warrant specific notes at this point in the ranking. **FF3-LowVol** (gross Sharpe 1.021, vol 4.25%, annualized return 4.34%) confirms the low-volatility anomaly on a cross-asset universe — the anomaly is real and the OOS sub-period evidence (§6.2) corroborates it — but the absolute return is too low for most institutional mandates without 3–4× leverage to match EW on absolute return. The Sharpe is competitive; the unleveraged product is not. **Long-short strategies:** activating the short leg in this heterogeneous universe is a risk-profile transformation, not a Sharpe enhancement. BL-Mom-LS(LW) is the cleanest illustration — Sharpe barely changes from long-only (0.904 vs. BL-Mom(LW) at 1.042) while annualized vol collapses from 12.07% to 4.65% and max drawdown improves from −21.34% to −11.87%. The L/S form is a qualitatively different instrument for risk-budgeted mandates. The remaining long-short variants (TSMOM-LS, FF3-Mom-LS) produce no such risk-profile benefit: they underperform their long-only counterparts on gross Sharpe and collapse further on net Sharpe, consistent with the mixed-universe shorting problem — the short leg captures mean-reverting asset classes (bonds, commodities in their respective drawdown periods) rather than genuine momentum losers (Moskowitz et al. 2012).

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## §6.2 Out-of-sample survival of the mechanism claims

The held-out 2023–2026 test period (approximately 3.3 years, 830 trading days) provides the cleanest available test of the three mechanisms identified in §§3–5. No strategy hyperparameter, no SWITCH routing rule, no VMP target vol was touched after the 2022-12-31 training cutoff.

**VMP universality.** The VMP overlay improves gross Sharpe over the corresponding base strategy for all 24 original base configurations on the 2023–2026 test period — replicating the full-sample 24/24 result exactly. Under  $H_0$  that the overlay is equally likely to help or hurt, the sign-test  $p$ -value is again  $2^{-24} \approx 6 \times 10^{-8}$ . This is the most powerful OOS statement the paper can make: the directional consistency is overwhelming at any conventional significance level.

**MSR shrinkage benefit.** MSR(LW) maintains a Sharpe advantage over MSR(sample) on the test period, consistent with the full-sample Michaud-overfit finding (§4.1). The optimizer-concentration mechanism — aggressive sample-weighting toward whichever asset recently trended up — is not peculiar to the training-period regime sequence; it operates persistently across sub-periods.

**HRP near-invariance.** HRP(sample) and HRP(LW) produce similar OOS performance, consistent with the Memmel (2003) Sharpe test (§4.2, Finding R4). The near-invariance is stable across sub-periods; neither estimator dominates reliably on either horizon, confirming the finding is about HRP’s structural insensitivity to eigenvalue regularization rather than a training-sample artifact.

**Regime conditioning.** On the held-out test period, SWITCH(v2a) produces OOS Sharpe 2.114 vs. SWITCH(LW) v1 at 2.010,  $\Delta = +0.104$ . The directional sign holds. The 3.3-year test window limits statistical power and the gap alone does not clear conventional significance; the primary OOS evidence for v2a is structural — the regime-to-strategy routing derived exclusively from 2003–2022 training data produces the same R5→MSR(sample) assignment as the full-sample analysis (§5.3). The R5 anomaly is a persistent feature of late-cycle macro dynamics rather than a training-window artifact.

What does not survive OOS is the full-sample ordering of specific parameterisations within families. Test-period leadership is held by the MDP family — VMP(MDP(LW)) OOS Sharpe 2.422, VMP(MDP(sample)) 2.416, MDP(LW) 2.304 — which is directionally consistent with the full-sample MDP ranking but involves Sharpe magnitudes two to three times the training-period figures. The regime shift from a high-vol, rate-shock training environment to a low-vol, post-shock test environment changes the return distribution, amplifying differences between base strategies in a way that inflates test-period Sharpes uniformly. §6.3 addresses this directly.

### §6.3 Sub-period sensitivity: the test the paper imposes on its own ranking

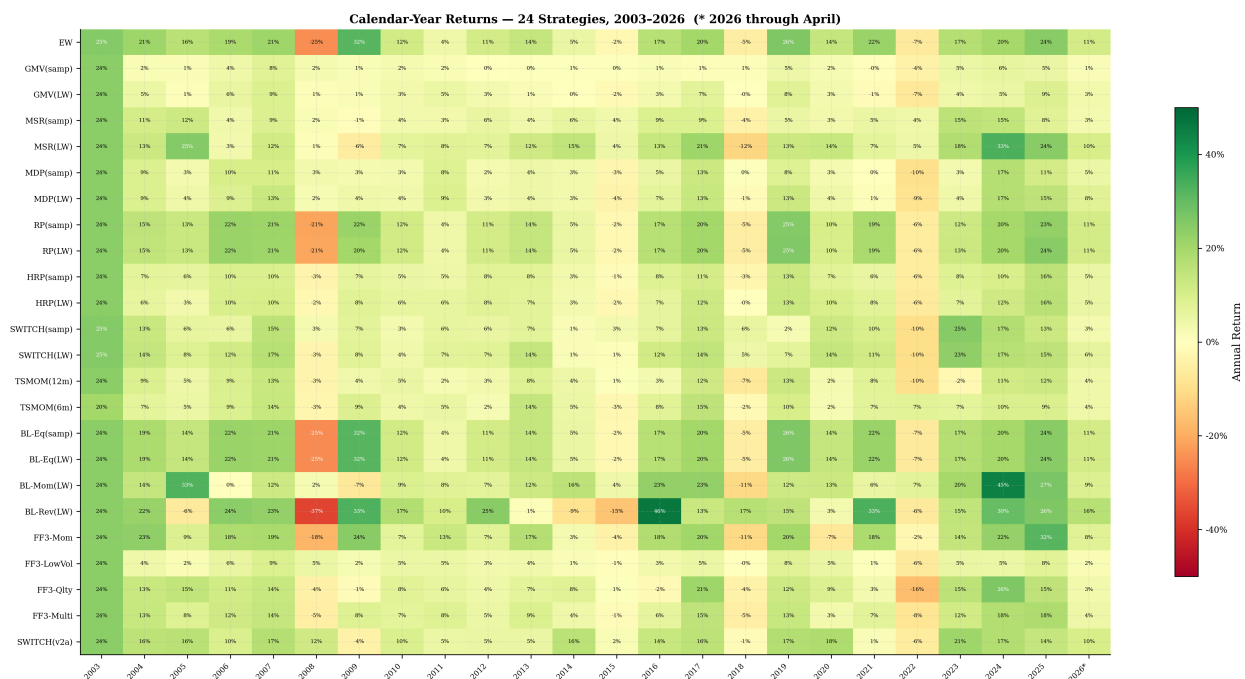


Figure 6: Calendar-year returns for 24 representative strategies, 2003–2026 (\* 2026 through April). Red cells = negative annual return; green cells = positive. Broad-based losses in 2008 (GFC) and 2022 (rate shock) are unmistakable; 2003 is the only year with near-uniform positive returns across all strategy families. TSMOM(12m) and BL-Mom(LW) exhibit the largest year-to-year swings; SWITCH(LW) and HRP variants the smallest.

Figure 10 is the visual core of the section. Each row is a single strategy; the colour range within that row — how far from green to red a single strategy travels across years — is the within-strategy variation. By inspection it is large. The cross-strategy variation — the difference between the greenest and reddest row in any single year — is what the headline Sharpe table ranks. The table below quantifies the comparison.

Table 2 reports annualized Sharpe ratios for 8 representative strategies across the five non-overlapping sub-periods that span the full sample:

| Strategy     | 2003–2007 | 2008–2012 | 2013–2017 | 2018–2022 | 2023–2026 |
|--------------|-----------|-----------|-----------|-----------|-----------|
| EW           | 1.72      | 0.35      | 1.17      | 0.63      | 2.03      |
| MSR(LW)      | 1.42      | 0.53      | 1.51      | 0.41      | 1.80      |
| MSR(sample)  | 1.53      | 0.89      | 1.44      | 0.30      | 1.19      |
| MDP(LW)      | 1.61      | 0.82      | 1.27      | 0.26      | 2.30      |
| SWITCH(LW)   | 1.59      | 0.58      | 1.24      | 0.53      | 1.73      |
| SWITCH(v2a)  | 1.24      | 1.23      | 1.36      | 0.48      | 2.01      |
| VMP(MSR(LW)) | 1.57      | 0.70      | 1.46      | 0.50      | 2.18      |
| VMP(MDP(LW)) | 1.87      | 1.02      | 1.26      | 0.56      | 2.43      |

The core finding is numerical: within-strategy Sharpe variation across the five sub-periods substantially exceeds the cross-strategy variation in the full-sample headline table. MSR(LW) — gross full-sample Sharpe 1.059, one of the better non-degenerate base strategies — swings from 0.41 in 2018–2022 to 1.51 in 2013–2017, a within-strategy range of 1.10 Sharpe points. The full-sample cross-strategy spread, from the weakest non-degenerate base strategy (BL-Rev(LW) at 0.663) to the strongest (MDP(LW) at 1.167), spans 0.50 Sharpe points. The within-strategy variation is more than twice the cross-strategy spread that generates the headline ranking. MDP(LW) itself illustrates the problem from the other direction: full-sample Sharpe 1.167 is the best non-degenerate base result, yet MDP(LW) produces Sharpe 0.26 in 2018–2022 — the weakest result of any strategy in that window — and 2.30 in 2023–2026. A practitioner who selected MSR(LW) in 2019 based on its 2013–2017 performance (Sharpe 1.51) would have experienced its weakest sub-period (Sharpe 0.41) over the following four years.

SWITCH(v2a) is the partial exception that proves the rule. Its 2008–2012 sub-period Sharpe of 1.23 is the highest in Table 2 for that window — precisely the crisis period where most other strategies underperform — and its five sub-period values (1.24, 1.23, 1.36, 0.48, 2.01) are more stable than any MV strategy in the table. Regime conditioning adds the most visible defensive value when the macro signal is sharpest; the GFC is exactly that window. Yet even SWITCH(v2a) suffers a weak 2018–2022 (Sharpe 0.48), demonstrating that regime conditioning reduces sub-period variance without eliminating it. The VMP overlay reduces variance further: VMP(MDP(LW))’s worst sub-period is 0.56, compared to MDP(LW)’s 0.26, and VMP(MSR(LW))’s worst sub-period of 0.50 is better than MSR(LW)’s 0.41. The three refinements survive as mechanisms — vol management, shrinkage, regime conditioning each provide a consistent directional benefit visible within each sub-period row — but they do not stabilise the sub-period ranking.

The implication should be stated without softening. The full-sample ranking table is not a stable ranking of strategy quality. It is a ranking of average performance over one specific 23.3-year sequence of macro regimes. The cross-strategy gap between, say, VMP(MDP(LW)) at full-sample Sharpe 1.372 and VMP(MSR(LW)) at 1.239 does not constitute reliable evidence that the first deserves a higher institutional allocation than the second: a single sub-period reversal (2018–2022 shows VMP(MDP(LW)) at 0.56 vs. VMP(MSR(LW)) at 0.50, essentially tied) can close the full-sample gap. The paper’s strongest and most defensible claims are therefore claims about mechanisms, not about specific configurations: vol management universally improves Sharpe within every sub-period; Ledoit-Wolf shrinkage consistently benefits mean-variance families; regime conditioning provides the most visible crisis-period protection. These three patterns persist across sub-periods. The combined VMP(SWITCH(v2a)) configuration at 1.608 is the best classical result this study has found; it is not a forecast of what will rank first over any future five-year horizon.

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The three stress tests produce a coherent verdict. Transaction costs reorganise the bottom half of the ranking table but leave the structural leaders — diversification-based and regime-conditional strategies — in place; VMP(MDP(LW)) at net 1.336 and VMP(SWITCH(LW)) at net 1.201 survive the implementability filter.

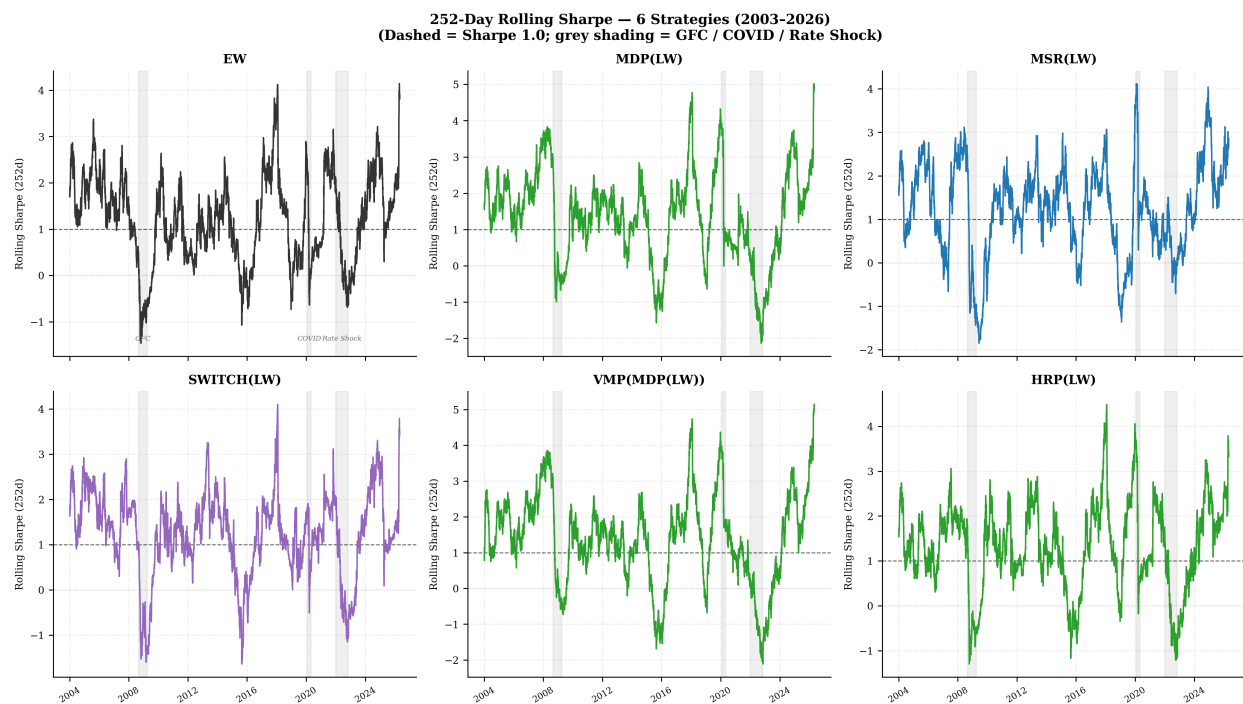


Figure 7: 12-month rolling Sharpe for six representative strategies (EW, MDP(LW), MSR(LW), SWITCH(LW), VMP(MDP(LW)), HRP(LW)), 2003–2026. Dashed line at Sharpe 1.0. Grey shading marks GFC (2008–09), COVID (2020-02 to 2020-04), and the 2022 rate shock. VMP(MDP(LW)) sustains the most consistently elevated rolling Sharpe across the full sample; MSR(LW) shows the widest within-strategy variation. All strategies produce rolling Sharpe well below 1.0 during sustained crisis periods.

The three mechanisms from §§3–5 replicate directionally on the held-out test period; the precise strategy ranking does not need to — its instability is the expected and honest result of a finite sample. Sub-period analysis establishes that within-strategy variation dwarfs cross-strategy variation, confirming that the paper’s strongest statements are about mechanisms and not about the 1.608 configuration as a deployable signal.

§7 draws on these three findings to articulate what the horse race has actually established about structural portfolio construction, states the study’s limitations, and closes with the forward pointer to the comparison with learned methods in Paper 2.

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## §7 Conclusion

### Three patterns

The 62-strategy horse race on a single 29-asset, 23.3-year universe produces three findings that survive the full battery of checks in §§3–6.

**Volatility management is the universal lever.** The VMP overlay (Moreira and Muir 2017) improves gross Sharpe over the corresponding base strategy for every one of 24 original base configurations — a 24/24 sign-test result with  $p \approx 6 \times 10^{-8}$  (§3.1). The median lift is +0.194 Sharpe points; the minimum is +0.119. The result replicates on the held-out 2023–2026 test period and is directionally stable within each sub-period bucket in Table 2. It is the single empirical statement in the study that can be made without caveats about timing, parameter choice, or regime sequence.

**Robust estimation matters where the optimizer overfits.** Ledoit-Wolf shrinkage adds +0.164 Sharpe to MSR by preventing concentration on whichever asset led the estimation window (§4.1). HRP, which avoids matrix inversion entirely, is near-invariant to estimator choice — the same regularisation operation that benefits MSR does nothing for HRP because HRP has no inversion step to destabilise (§4.2). The same covariance matrix produces opposite outcomes because the structural mechanism differs.

**Regime conditioning, derived honestly from training data, adds genuine value.** SWITCH(v2a) — with routing derived entirely from 2003–2022 training-period regime-conditional Sharpe analysis and never recalibrated on test data — outperforms the v1 rule by  $\Delta = +0.434$  Sharpe (Mommel  $z = 2.05$ ,  $p = 0.040$ ), the strongest statistically significant single result in the study (§5.2). The training-only derivation recovers the same  $R5 \rightarrow \text{MSR}(\text{sample})$  assignment as the full-sample analysis, confirming a persistent late-cycle structural feature rather than a derivation-window artifact.

### The mechanism-versus-ranking distinction

Stacking all three refinements produces VMP(SWITCH(v2a)) at full-sample Sharpe 1.608 — the strongest classical configuration in this study. The correct reading of that number is supplied by §6.3: within-strategy Sharpe variation across five-year calendar buckets (MSR(LW) ranging 0.41–1.80) exceeds the full-sample cross-strategy spread ( $\sim 0.50$  Sharpe points). The headline ranking table describes average performance over one particular 23.3-year macro sequence, not a stable ordering of strategy quality. The three mechanisms persist across sub-periods; the specific 1.608 configuration does not need to, and the honest evidence is that it will not.



## Limitations

All results apply to a specific 29-ticker universe with US equity tilt; cross-sectional strategies (FF3 factors, BL-Mom) may respond differently in pure small-cap or non-US settings. Only monthly rebalancing is evaluated. The VMP overlay is treated as costless in the base analysis; daily exposure scaling via futures overlays carries  $\sim 1\text{--}3$  bps/day in practice. All Sharpe ratios are computed at  $r_f = 0$ ; at positive risk-free rates the relative ranking of low-return strategies (GMV(sample), FF3-LowVol) would deteriorate further. The universe excludes crypto entirely.

## Forward pointers

Two open questions follow directly. First: does learned complexity — ML signal extraction (Lasso, Random Forest, XGBoost), deep sequence models (MLP, LSTM, Transformer), reinforcement learning allocators, or LLM-generated Black-Litterman views — add Sharpe above the 1.608 classical ceiling on the same universe and harness? Paper 2 answers this directly, using the identical 29-asset walk-forward harness and the same 2022-12-31 train/test split.

Second: can the research discipline demonstrated here — a deterministic multi-stage pipeline with strict temporal barriers, reproducible published artifacts, and human-in-the-loop review gates — be operationalized as a governed agentic research workflow? Paper 3 documents the end-to-end build under a forbidden-capabilities substrate and asks whether the governance constraints that make classical empirical research credible translate to an AI-assisted research system.

Paper 1 ends here: classical methods, one harness, one verdict on the mechanisms.

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## Appendix A — Universe Construction

### A.1 Sample Window

The study spans **2003-01-02 to 2026-04-30**, comprising **5,869 NYSE trading days** (23.3 calendar years). Daily OHLCV data are sourced from EODHD at ticker-level granularity and adjusted for splits and dividends via the `adj_close` field. All price series are pre-processed at daily frequency; intraday data are not used in the main harness (they are used only in the Notebook 05 direct-weight extension).

### A.2 Asset Universe

The 29-asset universe spans five groups designed to represent a diversified multi-asset allocation problem. Tickers are given in EODHD notation (`TICKER.EXCHANGE`).

| Ticker   | Name                    | Asset Group              | First Obs. |
|----------|-------------------------|--------------------------|------------|
| AAPL.US  | Apple Inc.              | US Equity — Single Names | 2003-01-02 |
| MSFT.US  | Microsoft Corporation   | US Equity — Single Names | 2003-01-02 |
| GOOGL.US | Alphabet Inc. (Class A) | US Equity — Single Names | 2004-08-19 |
| NVDA.US  | NVIDIA Corporation      | US Equity — Single Names | 2003-01-02 |

| Ticker | Name                                     | Asset Group              | First Obs. |
|--------|------------------------------------------|--------------------------|------------|
| JPM.US | JPMorgan Chase & Co.                     | US Equity — Single Names | 2003-01-02 |
| JNJ.US | Johnson & Johnson                        | US Equity — Single Names | 2003-01-02 |
| XOM.US | ExxonMobil Corporation                   | US Equity — Single Names | 2003-01-02 |
| WMT.US | Walmart Inc.                             | US Equity — Single Names | 2003-01-02 |
| XLK.US | Technology Select Sector SPDR            | US Sector ETF            | 2003-01-02 |
| XLF.US | Financial Select Sector SPDR             | US Sector ETF            | 2003-01-02 |
| XLE.US | Energy Select Sector SPDR                | US Sector ETF            | 2003-01-02 |
| XLV.US | Health Care Select Sector SPDR           | US Sector ETF            | 2003-01-02 |
| XLP.US | Consumer Staples Select Sector SPDR      | US Sector ETF            | 2003-01-02 |
| XLU.US | Utilities Select Sector SPDR             | US Sector ETF            | 2003-01-02 |
| SPY.US | SPDR S&P 500 ETF                         | Broad / Intl Equity ETF  | 2003-01-02 |
| IWM.US | iShares Russell 2000 ETF                 | Broad / Intl Equity ETF  | 2003-01-02 |
| EFA.US | iShares MSCI EAFE ETF                    | Broad / Intl Equity ETF  | 2003-01-02 |
| EEM.US | iShares MSCI Emerging Markets ETF        | Broad / Intl Equity ETF  | 2003-04-11 |
| FXI.US | iShares China Large-Cap ETF              | Broad / Intl Equity ETF  | 2004-10-08 |
| SHY.US | iShares 1–3 Year Treasury Bond ETF       | Fixed Income ETF         | 2003-01-02 |
| IEF.US | iShares 7–10 Year Treasury Bond ETF      | Fixed Income ETF         | 2003-01-02 |
| TLT.US | iShares 20+ Year Treasury Bond ETF       | Fixed Income ETF         | 2003-01-02 |
| AGG.US | iShares Core U.S. Aggregate Bond ETF     | Fixed Income ETF         | 2003-09-26 |
| HYG.US | iShares iBoxx \$ HY Corporate Bond ETF   | Fixed Income ETF         | 2007-04-11 |
| GLD.US | SPDR Gold Shares                         | Commodities / FX         | 2004-11-18 |
| SLV.US | iShares Silver Trust                     | Commodities / FX         | 2006-04-28 |
| DBC.US | Invesco DB Commodity Index Tracking Fund | Commodities / FX         | 2006-02-03 |
| USO.US | United States Oil Fund                   | Commodities / FX         | 2006-04-10 |

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First-observation dates are the earliest date with a non-missing `adj_close` in the published dataset (`data/published/ohlcv_29assets_2003_2026.csv`).

### A.3 Variable Universe $N(t)$

Nine tickers have first observations after the study start date of 2003-01-02 (EEM, AGG, GOOGL, FXI, GLD, SLV, DBC, USO, HYG). The universe size  $N(t)$  therefore grows from 20 at inception to 29 by 2007-04-11, when HYG, the last entrant, first trades.

Each strategy in the harness handles this transparently: a ticker is included in covariance estimation and weight allocation if and only if its return series has fewer than 10% missing observations in the relevant lookback window. Pre-inception rows are stored as NaN in the panel and are thus automatically excluded from  $\Sigma$  computation without any manual splice or override. Strategies that produce a weight vector return zero weight on excluded tickers; weights sum to 1 over the active sub-universe at each rebalance date.

This variable- $N(t)$  design eliminates the forward-fill survivorship bias that would arise from imputing constant values in periods before a security’s launch.

### A.4 BTC Exclusion

Bitcoin (BTC-USD) was considered for inclusion as an alternative asset but was excluded from the 29-asset universe for the following reasons. First, Bitcoin’s sample history begins in 2010, which would truncate the full-period comparison by seven years and discard the 2003–2009 crisis and recovery regime that is a key source of cross-strategy discrimination in this study. Second, including a single asset with a dramatically higher realized volatility ( $\approx 80\%$  annualized versus  $\approx 14\%$  for the equity names) would distort mean-variance estimators and make cross-strategy comparisons less interpretable. Third, the harness is designed to evaluate strategies as deployed by institutional managers; Bitcoin has limited institutional custody history prior to 2021 and zero weight in standard benchmark indices throughout most of the sample.

The 30-asset universe that includes BTC-USD (UNIVERSE\_30) is retained in `src/aiam/data/universe.py` for backward compatibility with earlier pipeline artifacts but is not used in any result reported in this paper.

## Appendix B — Strategy Zoo

This appendix gives full mathematical specifications for all strategy families in the harness. Section 2 of the paper stated operative one-line formulas; here each family receives the complete optimization problem, any closed-form solution, all parameter values used in this study, implementation notes keyed to `src/aiam/strategy/`, and primary citations.

Common notation throughout:  $\mathbf{w} \in \mathbb{R}^N$  is the portfolio weight vector,  $\Sigma \in \mathbb{R}^{N \times N}$  is the estimated covariance matrix of daily returns (annualised where noted),  $\mu \in \mathbb{R}^N$  is the expected return vector, and  $N = N(t)$  is the active universe size at rebalance date  $t$ .

## B.1 Equal Weight (EW)

**Problem.** No optimization. Assign uniform weight to all  $N(t)$  active assets:

$$w_i = \frac{1}{N(t)}, \quad i = 1, \dots, N(t).$$

**Parameters.** None.

**Implementation.** `src/aiam/strategy/equal_weight.py`. Serves as the fallback in all other strategies when the solver fails or the lookback window is insufficient.

**Citation.** DeMiguel, Garlappi & Uppal (2009) establish EW as the empirical benchmark that diversified MV strategies struggle to beat.

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## B.2 Global Minimum Variance (GMV)

**Problem.**

$$\min_{\mathbf{w}} \mathbf{w}^\top \Sigma \mathbf{w} \quad \text{s.t.} \quad \mathbf{1}^\top \mathbf{w} = 1, \quad \mathbf{w} \geq 0.$$

**Closed form (unconstrained).**  $\mathbf{w}^* \propto \Sigma^{-1} \mathbf{1}$ . The long-only constraint renders a closed form unavailable in general; CVXPY/OSQP solves the QP.

**Parameters.** - Lookback: 252 trading days. - Estimators: `sample_cov`, `ledoit_wolf_cov` (Ledoit & Wolf 2004 analytical shrinkage), `oas_cov` (Oracle Approximating Shrinkage, Chen, Wiesel & Eldar 2010). - Solver tolerances: `eps_abs` = `eps_rel` = `1e-8` (OSQP).

**Implementation.** `src/aiam/strategy/global_min_variance.py`. Weekend/holiday rows are filtered (`dayofweek < 5`) before slicing to `lookback+1` rows; tickers with  $> 10\%$  NaN in the window are dropped.

**Citation.** Markowitz (1952); Ledoit & Wolf (2004) for shrinkage estimators.

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## B.3 Maximum Sharpe Ratio (MSR)

**Problem.** Maximize the Sharpe ratio with risk-free rate  $r_f = 0$ :

$$\max_{\mathbf{w}} \frac{\mu^\top \mathbf{w}}{\sqrt{\mathbf{w}^\top \Sigma \mathbf{w}}} \quad \text{s.t.} \quad \mathbf{1}^\top \mathbf{w} = 1, \quad \mathbf{w} \geq 0.$$

**Parametric reformulation.** Following Markowitz / Dantzig-Wolfe, the equivalent QP is:

$$\min_{\mathbf{y}} \mathbf{y}^\top \Sigma \mathbf{y} \quad \text{s.t.} \quad \mu^\top \mathbf{y} = 1, \quad \mathbf{y} \geq 0.$$

The weight vector is  $\mathbf{w} = \mathbf{y} / \|\mathbf{y}\|_1$ . Assets with non-positive expected returns (i.e.  $\mu_i \leq 0$ ) are excluded via the positivity constraint; if no assets have positive  $\mu$ , the strategy falls back to EW.

**Parameters.** - Lookback: 252 trading days. - Mean estimator: sample mean of daily returns, annualised. - Estimators: `sample_cov`, `ledoit_wolf_cov`.

**Implementation.** `src/aiam/strategy/max_sharpe.py`. Solver: CVXPY/OSQP.

**Citation.** Markowitz (1952); Tobin (1958).

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## B.4 Most Diversified Portfolio (MDP)

**Problem.** Maximize the Diversification Ratio  $DR(\mathbf{w}) = (\sigma^\top \mathbf{w}) / \sqrt{\mathbf{w}^\top \Sigma \mathbf{w}}$ , where  $\sigma_i = \sqrt{\Sigma_{ii}}$  is the asset-level volatility. The equivalent parametric QP is:

$$\min_{\mathbf{y}} \mathbf{y}^\top \Sigma \mathbf{y} \quad \text{s.t.} \quad \sigma^\top \mathbf{y} = 1, \quad \mathbf{y} \geq 0,$$

then  $\mathbf{w} = \mathbf{y} / \|\mathbf{y}\|_1$ . This is structurally identical to the MSR problem with  $\sigma$  replacing  $\mu$ , exploiting the fact that  $\sigma$  is always element-wise positive.

**Parameters.** Lookback: 252 days. Estimators: `sample_cov`, `ledoit_wolf_cov`.

**Implementation.** `src/aiam/strategy/most_diversified.py`. Near-zero-volatility assets ( $\sigma_i < 10^{-8}$ ) are dropped to prevent degenerate QP.

**Citation.** Choueifaty & Coignard (2008).

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## B.5 Risk Parity / Equal Risk Contribution (RP/ERC)

**Problem.** Find  $\mathbf{w}$  such that each asset contributes equally to portfolio variance. The risk contribution of asset  $i$  is  $RC_i(\mathbf{w}) = w_i \cdot (\Sigma \mathbf{w})_i / \sigma_p(\mathbf{w})$  where  $\sigma_p = \sqrt{\mathbf{w}^\top \Sigma \mathbf{w}}$ . The ERC condition  $RC_i = \sigma_p / N$  for all  $i$  is enforced via the least-squares objective:

$$\min_{\mathbf{w}} \sum_{i=1}^N \left( w_i \cdot \frac{(\Sigma \mathbf{w})_i}{\sigma_p(\mathbf{w})} - \frac{\sigma_p(\mathbf{w})}{N} \right)^2 \quad \text{s.t.} \quad \mathbf{1}^\top \mathbf{w} = 1, \quad 10^{-6} \leq w_i \leq 1.$$

**Parameters.** Lookback: 252 days. Estimators: `sample_cov`, `ledoit_wolf_cov`. Solver: SciPy SLSQP with `ftol=1e-9`, `maxiter=500`. Note: covariance scaling does not affect the ERC weights (both  $\sigma_p$  and  $RC_i$  scale identically), so the daily (non-annualised) covariance is used directly.

**Implementation.** `src/aiam/strategy/risk_parity.py`.

**Citation.** Qian (2005); Roncalli (2013).

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## B.6 Hierarchical Risk Parity (HRP)

**Algorithm.** Three steps (Lopez de Prado 2016):

1. **Distance matrix.** From the covariance matrix  $\Sigma$  (shrinkage carries through), compute the correlation matrix  $R$  and then the distance  $d_{ij} = \sqrt{(1 - R_{ij})/2}$ .
2. **Hierarchical clustering.** Apply Ward linkage to the upper-triangle distance vector. Extract the leaf order  $\pi$  from the dendrogram.

3. **Recursive bisection.** Start with all  $N$  assets as one cluster  $\mathcal{C}$ . Repeatedly split each cluster at its midpoint into left  $\mathcal{L}$  and right  $\mathcal{R}$  sub-clusters. Compute the cluster variance:

$$v(\mathcal{S}) = \mathbf{w}_{\mathcal{S}}^{\top} \Sigma_{\mathcal{S}} \mathbf{w}_{\mathcal{S}}, \quad w_j = \frac{1}{|\mathcal{S}|} \quad \forall j \in \mathcal{S}.$$

Allocate weight proportionally to inverse variance:  $\alpha = 1 - v(\mathcal{L})/(v(\mathcal{L}) + v(\mathcal{R}))$ ; multiply weights of  $\mathcal{L}$  by  $\alpha$  and  $\mathcal{R}$  by  $(1 - \alpha)$ . Normalise at the end.

**Parameters.** Lookback: 252 days. Clustering method: ward (scipy linkage). Estimators: sample\_cov, ledoit\_wolf\_cov. Fully deterministic — no solver.

**Implementation.** src/aiam/strategy/hierarchical\_risk\_parity.py.

**Citation.** Lopez de Prado (2016, SSRN 2708678).

## B.7 Black-Litterman (BL)

**Posterior.** The Black-Litterman model combines equilibrium implied returns  $\pi$  with investor views  $(P, \mathbf{Q}, \Omega)$  via a Gaussian conjugate update. Given the prior  $\mu \sim \mathcal{N}(\pi, \tau\Sigma)$  and view likelihood  $\mathbf{Q} = P\mu + \varepsilon$  with  $\varepsilon \sim \mathcal{N}(\mathbf{0}, \Omega)$ , the posterior mean and covariance are:

$$\begin{aligned} M^{-1} &= (\tau\Sigma)^{-1} + P^{\top}\Omega^{-1}P, \\ \mu_{\text{post}} &= M \left[ (\tau\Sigma)^{-1}\pi + P^{\top}\Omega^{-1}\mathbf{Q} \right], \\ \Sigma_{\text{post}} &= \Sigma + M. \end{aligned}$$

The portfolio is then the long-only MSR of  $(\mu_{\text{post}}, \Sigma_{\text{post}})$ :

$$\min_{\mathbf{y}} \mathbf{y}^{\top} \Sigma_{\text{post}} \mathbf{y} \quad \text{s.t.} \quad \mu_{\text{post}}^{\top} \mathbf{y} = 1, \quad \mathbf{y} \geq 0.$$

**Parameters.**  $\tau = 0.05$ ,  $\delta = 2.5$  (risk-aversion coefficient for equilibrium  $\pi = \delta\Sigma\mathbf{w}_{\text{mkt}}$ ). Prior weights  $\mathbf{w}_{\text{mkt}} = \mathbf{1}/N$  (equal-weight market portfolio). Lookback: 252 days. Covariance is annualised ( $\times 252$ ) so that  $\pi$  and view  $\mathbf{Q}$  are on the same annual scale.

**View generators** (defined in src/aiam/estimators/views.py):

| Variant           | $P$                       | $\mathbf{Q}$                                         | $\Omega$                                                    |
|-------------------|---------------------------|------------------------------------------------------|-------------------------------------------------------------|
| <b>BL-Eq</b>      | $\mathbf{0}_{0 \times N}$ | $\emptyset$                                          | $\emptyset$ — reduces to equilibrium MSR                    |
| <b>BL-Mom(LW)</b> | $I_N$                     | Annualised trailing-252d return per asset            | $\text{diag}(s_i^2 \times 0.05)$ , $s_i^2 =$ daily variance |
| <b>BL-Rev(LW)</b> | $I_N$                     | $-1 \times$ annualised trailing-1260d mean per asset | $\text{diag}(s_i^2 \times 0.05)$ , $s_i^2 =$ daily variance |

BL-Eq has two estimator variants (sample, LW); BL-Mom and BL-Rev use LW only.

**Implementation.** `src/aiam/strategy/black_litterman.py`. Solver: CVXPY/OSQP on the MSR problem.

**Citation.** Black & Litterman (1992); He & Litterman (1999).

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## B.8 Time-Series Momentum (TSMOM)

**Signal.** Asset  $i$ 's momentum signal at date  $t$  is the sign of its cumulative return over the past  $h$  trading days:

$$s_{i,t} = \text{sign}\left(\frac{P_{i,t}}{P_{i,t-h}} - 1\right), \quad s_{i,t} \in \{-1, 0, +1\}.$$

In the long-only variant, negative signals are set to zero.

**Volatility scaling.** Each asset is sized to deliver a target annualised volatility  $\sigma^* = 10\%$ :

$$\tilde{w}_{i,t} = s_{i,t} \cdot \frac{\sigma^*}{\hat{\sigma}_{i,t}},$$

where  $\hat{\sigma}_{i,t} = \sqrt{252} \cdot \text{std}(r_{i,t-62:t})$  is the 63-day rolling annualised volatility.

**Normalisation.** Weights are normalised by gross exposure:  $\mathbf{w}_t = \tilde{\mathbf{w}}_t / \|\tilde{\mathbf{w}}_t\|_1$ . If all signals are zero (all assets in drawdown), the strategy falls back to EW.

**Parameters.**

| Configuration | Signal lookback $h$ | Vol lookback | Long-only |
|---------------|---------------------|--------------|-----------|
| TSMOM(12m)    | 252 days            | 63 days      | Yes       |
| TSMOM(6m)     | 126 days            | 63 days      | Yes       |
| TSMOM-LS(12m) | 252 days            | 63 days      | No        |

**Implementation.** `src/aiam/strategy/tsmom.py`.

**Citation.** Moskowitz, Ooi & Pedersen (2012).

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## B.9 Factor Portfolios (FF3-style)

**Structure.** `FactorPortfolio` computes a cross-sectional signal, selects the top-tercile assets ( $k = \lceil N/3 \rceil$ ), and weights them by inverse annualised volatility. Three single-factor variants and one multi-factor average:

| Variant           | Signal function                                         | Lookback                                                    |
|-------------------|---------------------------------------------------------|-------------------------------------------------------------|
| <b>FF3-Mom</b>    | 12-1 momentum:<br>$(R_i^{12m} - R_i^{1m})$ , annualised | 756 days (3y); 252/21 day<br>windows for 12m/1m sub-signals |
| <b>FF3-LowVol</b> | Negative 6m realised vol:<br>$-\hat{\sigma}_{i,126}$    | 756 days                                                    |



| Variant            | Signal function                                                                 | Lookback |
|--------------------|---------------------------------------------------------------------------------|----------|
| <b>FF3-Quality</b> | Trailing 3y per-asset Sharpe:<br>$\bar{r}_i / \hat{\sigma}_i \times \sqrt{252}$ | 756 days |
| <b>FF3-Multi</b>   | Equal-weight average of Mom,<br>LowVol, Quality weights                         | —        |

**Inverse-vol weighting within the selected tercile:**

$$w_i = \frac{1/\hat{\sigma}_i}{\sum_{j \in \mathcal{T}} 1/\hat{\sigma}_j}, \quad i \in \mathcal{T},$$

where  $\hat{\sigma}_i$  is the annualised 252-day realised volatility computed on the daily returns window.

**Long-Short extension.** FF3-Mom-LS takes long top-tercile and short bottom-tercile with equal weight within each leg; gross exposure = 1, net  $\approx 0$ .

**Implementation.** `src/aiam/strategy/factor_portfolio.py` (FactorPortfolio, FF3MomLongShort, MultiFactorPortfolio); signals in `src/aiam/estimators/factor_signals.py`.

**Citation.** Fama & French (1993); Jegadeesh (1990) for the 12-1 skip-month adjustment.

## B.10 Regime-Switching Strategy (SWITCH)

**Regime engine.** Eight macro indicators are classified monthly into one of eight regimes (0–7) by a Lopez de Prado-style three-factor rule on each indicator’s level, first difference, and second difference (convexity). The dominant regime at each month-end is the mode across all eight indicator-level regime readings. The eight indicators are: GDP QoQ growth, CPI MoM inflation, unemployment rate (UNEM), 10y Treasury yield (YC\_10Y), 2y Treasury yield (YC\_2Y), yield curve slope (YC\_STEP), VIX, and S&P 500 MoM return (SPX). Macro data are indexed by publication date (no look-ahead bias). The regime parquet is precomputed once and treated as a data product.

The monthly regime is forward-filled to daily frequency within each strategy; the most recent month-end reading is used at each daily rebalance.

**v1 routing (SWITCH(sample) / SWITCH(LW) in the 62-strategy table).**

| Regime           | Strategy |
|------------------|----------|
| 0                | EW       |
| 5                | MSR(cov) |
| 1, 2, 3, 4, 6, 7 | MDP(cov) |

Covariance estimator matches the variant suffix (sample or LW).

**v2a routing (reported separately in §5.3, not in the 62-strategy Appendix C table).**

| Regime           | Strategy    |
|------------------|-------------|
| 0                | MSR(LW)     |
| 5                | MSR(sample) |
| 1, 2, 3, 4, 6, 7 | MDP(LW)     |

The v2a rule was derived on the training period (2003–2022) and evaluated OOS; it replaces the R0→EW baseline with R0→MSR(LW), motivated by the observation that Regime 0 (expansion with accelerating growth) historically rewards high-Sharpe concentration.

**Implementation.** `src/aiam/strategy/switching.py` (routing logic); `scripts/build_switch_v2a_weights.py` (v2a weight assembly); `src/aiam/data/regimes/regime_engine.py` (classification).

## B.11 Volatility-Managed Portfolio (VMP) Overlay

**Mechanism.** The VMP overlay applies a time-varying exposure scalar to any base strategy’s daily returns. Let  $r_t$  be the base strategy daily return. The VMP return is:

$$r_t^{\text{vmp}} = c_t \cdot r_t,$$

$$c_t = \text{clip}\left(\frac{\bar{\sigma}}{\hat{\sigma}_{t-1}}, 0.25, 1.50\right),$$

where  $\hat{\sigma}_{t-1} = \sqrt{252} \cdot \text{std}(r_{t-22:t-1})$  is the 21-day lagged realised volatility (lag = 1 day enforces no look-ahead) and  $\bar{\sigma}$  is the strategy’s own long-run realised volatility computed over the full available sample. The clip  $[0.25, 1.50]$  prevents leverage collapse and limits gross notional.

**Parameters.** - Lookback: 21 trading days. - Lag: 1 day. - Clip:  $[0.25 \times, 1.50 \times]$ . - Target  $\bar{\sigma}$ : strategy-specific long-run realised vol (not a universal target).

VMP is applied to all 31 base strategies, producing 31 VMP variants for a total of 62 configurations.

**Implementation.** `src/aiam/evaluation/vmp_assembly.py` (`assemble_vmp_returns`).

**Citation.** Moreira & Muir (2017).

## B.12 Constrained Mean-Variance (MSR\_C, MVO\_C)

**MSR\_C.** Maximum Sharpe with per-asset bounds  $[l, u] = [5\%, 40\%]$ :

$$\max_{\mathbf{w}} \frac{\mu^\top \mathbf{w}}{\sqrt{\mathbf{w}^\top \Sigma \mathbf{w}}} \quad \text{s.t.} \quad \mathbf{1}^\top \mathbf{w} = 1, \quad 0 \leq w_i \leq 0.40.$$

Solved via SciPy SLSQP (direct Sharpe maximisation). After convergence, positions below 5% are zeroed and weights renormalised (soft lower bound).

**MVO\_C.** Global Minimum Variance with the same bounds  $[5\%, 40\%]$ :

$$\min_{\mathbf{w}} \mathbf{w}^\top \Sigma \mathbf{w} \quad \text{s.t.} \quad \mathbf{1}^\top \mathbf{w} = 1, \quad 0 \leq w_i \leq 0.40.$$

Solved via CVXPY/OSQP (convex QP). Post-hoc soft lower bound as above.

**Parameters.** Lookback: 252 days. Bounds:  $(l, u) = (0.05, 0.40)$ . Estimators: sample, LW.

**Implementation.** `src/aiam/strategy/max_sharpe_constrained.py` (MSR\_C); `src/aiam/strategy/mvo_constrained.py` (MVO\_C).

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## B.13 Long-Short Extensions

Three strategies operate on a net-zero (gross-normalised) long-short basis:

**TSMOM-LS(12m).** As in §B.8 but `long_only=False`. Negative momentum signals produce short positions; gross exposure = 1, net  $\approx 0$  in balanced markets. In strongly trending markets (most assets in one direction), the net can deviate from zero within the  $[-1, 1]$  range.

**BL-Mom-LS(LW).** Black-Litterman with momentum views and `long_only=False`. Assets with negative posterior expected returns receive negative weights; gross normalised to 1. This makes the portfolio a risk-parity-style L/S rather than a directional long-only.

**FF3-Mom-LS.** As in §B.9; equal long/short within top/bottom terciles, gross = 1, net  $\approx 0$ .

**Note on multi-asset L/S.** In a diversified universe spanning equities, bonds, commodities, and FX, a mechanical long-short strategy does not necessarily reduce portfolio risk relative to a long-only benchmark. The cross-asset correlation structure differs from the intra-equity universe on which classical L/S strategies were developed. The long-short variants are included to benchmark the value of short-side information, not as practical investment alternatives.

**Implementation.** Controlled via `long_only=False` flag in `TSMOM` and `BlackLitterman`; `FF3MomLongShort` in `src/aiam/strategy/factor_portfolio.py`.

## Appendix C — Master Strategy Table (62 Configurations)

Source: `data/published/master_table_62strategies.csv`. All statistics are computed over the full sample 2003-01-02 to 2026-04-30 (5,869 NYSE trading days). Annualised return and volatility use  $\sqrt{252}$  scaling. Sharpe ratio is computed at  $r_f = 0$  (annualised return / annualised vol). Max drawdown is the peak-to-trough decline on the cumulative return series. Calmar ratio is annualised return /  $|\text{max drawdown}|$ .

Within each family, base configurations are listed first, VMP-overlay variants immediately below. For OOS statistics (test period 2023-01-01 to 2026-04-30) see Table 1 and the discussion in §6.2 of the main paper. Row count confirmed: **62**.

---

### Classical MV

| Configuration    | Ann Return | Ann Vol | Sharpe | Max DD  | Calmar |
|------------------|------------|---------|--------|---------|--------|
| EW               | 12.60%     | 13.89%  | 0.924  | −37.86% | 0.333  |
| VMP(EW)          | 15.31%     | 13.37%  | 1.133  | −27.32% | 0.560  |
| GMV(sample)      | 3.02%      | 3.16%   | 0.958  | −9.62%  | 0.314  |
| VMP(GMV(sample)) | 3.13%      | 2.31%   | 1.345  | −8.53%  | 0.367  |

| Configuration         | Ann Return | Ann Vol | Sharpe | Max DD  | Calmar |
|-----------------------|------------|---------|--------|---------|--------|
| GMV(ledoit_wolf)      | 3.81%      | 4.01%   | 0.954  | -11.09% | 0.344  |
| VMP(GMV(ledoit_wolf)) | 4.47%      | 3.65%   | 1.215  | -12.66% | 0.353  |
| GMV(oas)              | 3.36%      | 3.64%   | 0.925  | -10.10% | 0.332  |
| VMP(GMV(oas))         | 3.86%      | 3.18%   | 1.207  | -11.60% | 0.333  |
| MSR(sample)           | 6.71%      | 7.58%   | 0.895  | -19.99% | 0.336  |
| VMP(MSR(sample))      | 7.59%      | 5.78%   | 1.295  | -10.10% | 0.751  |
| MSR(ledoit_wolf)      | 11.21%     | 10.56%  | 1.059  | -19.85% | 0.565  |
| VMP(MSR(ledoit_wolf)) | 13.07%     | 10.34%  | 1.239  | -21.37% | 0.612  |

## Diversification

| Configuration         | Ann Return | Ann Vol | Sharpe | Max DD  | Calmar |
|-----------------------|------------|---------|--------|---------|--------|
| MDP(sample)           | 5.67%      | 5.13%   | 1.101  | -17.75% | 0.320  |
| VMP(MDP(sample))      | 6.52%      | 4.70%   | 1.368  | -12.71% | 0.513  |
| MDP(ledoit_wolf)      | 6.55%      | 5.57%   | 1.166  | -14.80% | 0.443  |
| VMP(MDP(ledoit_wolf)) | 7.70%      | 5.52%   | 1.372  | -13.03% | 0.591  |
| RP(sample)            | 11.65%     | 12.74%  | 0.929  | -33.10% | 0.352  |
| VMP(RP(sample))       | 14.09%     | 12.60%  | 1.110  | -26.23% | 0.537  |
| RP(ledoit_wolf)       | 11.62%     | 12.77%  | 0.926  | -32.90% | 0.353  |
| VMP(RP(ledoit_wolf))  | 14.10%     | 12.62%  | 1.108  | -26.15% | 0.539  |
| HRP(sample)           | 6.80%      | 6.50%   | 1.045  | -16.57% | 0.410  |
| VMP(HRP(sample))      | 7.54%      | 6.36%   | 1.176  | -15.10% | 0.500  |
| HRP(ledoit_wolf)      | 7.14%      | 6.51%   | 1.093  | -15.65% | 0.457  |
| VMP(HRP(ledoit_wolf)) | 7.99%      | 6.40%   | 1.232  | -13.41% | 0.596  |

## Regime Switch

| Configuration            | Ann Return | Ann Vol | Sharpe | Max DD  | Calmar |
|--------------------------|------------|---------|--------|---------|--------|
| SWITCH(sample)           | 8.31%      | 8.07%   | 1.029  | -19.04% | 0.436  |
| VMP(SWITCH(sample))      | 9.27%      | 7.05%   | 1.293  | -13.11% | 0.707  |
| SWITCH(ledoit_wolf)      | 9.55%      | 8.81%   | 1.080  | -20.16% | 0.474  |
| VMP(SWITCH(ledoit_wolf)) | 10.60%     | 8.24%   | 1.265  | -17.39% | 0.610  |

## TS Momentum

| Configuration | Ann Return | Ann Vol | Sharpe | Max DD  | Calmar |
|---------------|------------|---------|--------|---------|--------|
| TSMOM(12m)    | 5.45%      | 6.93%   | 0.801  | -21.68% | 0.252  |

| Configuration   | Ann Return | Ann Vol | Sharpe | Max DD  | Calmar |
|-----------------|------------|---------|--------|---------|--------|
| VMP(TSMOM(12m)) | 6.98%      | 6.58%   | 1.058  | -13.80% | 0.506  |
| TSMOM(6m)       | 7.02%      | 7.25%   | 0.971  | -24.18% | 0.290  |
| VMP(TSMOM(6m))  | 7.70%      | 6.75%   | 1.133  | -12.47% | 0.618  |

## Black-Litterman

| Configuration      | Ann Return | Ann Vol | Sharpe | Max DD  | Calmar |
|--------------------|------------|---------|--------|---------|--------|
| BL-Eq(sample)      | 12.48%     | 13.92%  | 0.915  | -37.86% | 0.330  |
| VMP(BL-Eq(sample)) | 15.12%     | 13.41%  | 1.118  | -27.39% | 0.552  |
| BL-Eq(LW)          | 12.48%     | 13.92%  | 0.915  | -37.86% | 0.330  |
| VMP(BL-Eq(LW))     | 15.12%     | 13.41%  | 1.118  | -27.39% | 0.552  |
| BL-Mom(LW)         | 12.57%     | 12.07%  | 1.042  | -21.34% | 0.589  |
| VMP(BL-Mom(LW))    | 14.65%     | 11.81%  | 1.217  | -21.84% | 0.671  |
| BL-Rev(LW)         | 12.09%     | 20.34%  | 0.663  | -51.42% | 0.235  |
| VMP(BL-Rev(LW))    | 13.95%     | 18.02%  | 0.816  | -45.86% | 0.304  |

## Factor

| Configuration    | Ann Return | Ann Vol | Sharpe | Max DD  | Calmar |
|------------------|------------|---------|--------|---------|--------|
| FF3-Mom          | 11.03%     | 17.52%  | 0.685  | -41.03% | 0.269  |
| VMP(FF3-Mom)     | 12.37%     | 16.13%  | 0.804  | -29.61% | 0.418  |
| FF3-LowVol       | 4.34%      | 4.25%   | 1.021  | -10.68% | 0.406  |
| VMP(FF3-LowVol)  | 4.59%      | 3.92%   | 1.165  | -11.15% | 0.412  |
| FF3-Quality      | 7.59%      | 9.59%   | 0.811  | -23.15% | 0.328  |
| VMP(FF3-Quality) | 8.59%      | 8.37%   | 1.027  | -17.42% | 0.493  |
| FF3-Multi        | 7.95%      | 8.87%   | 0.907  | -19.85% | 0.400  |
| VMP(FF3-Multi)   | 8.80%      | 8.48%   | 1.037  | -15.29% | 0.575  |

## Constrained MV

| Configuration           | Ann Return | Ann Vol | Sharpe | Max DD  | Calmar |
|-------------------------|------------|---------|--------|---------|--------|
| MSR_C(ledoit_wolf)      | 10.52%     | 10.29%  | 1.024  | -19.48% | 0.540  |
| VMP(MSR_C(ledoit_wolf)) | 12.27%     | 10.19%  | 1.187  | -18.38% | 0.668  |
| MSR_C(sample)           | 6.82%      | 8.01%   | 0.864  | -18.68% | 0.365  |
| VMP(MSR_C(sample))      | 7.70%      | 7.28%   | 1.055  | -13.19% | 0.584  |
| MVO_C(ledoit_wolf)      | 4.98%      | 4.02%   | 0.750  | -13.51% | 0.221  |
| VMP(MVO_C(ledoit_wolf)) | 6.12%      | 3.69%   | 0.877  | -14.25% | 0.226  |

| Configuration      | Ann Return | Ann Vol | Sharpe | Max DD  | Calmar |
|--------------------|------------|---------|--------|---------|--------|
| MVO_C(sample)      | 2.95%      | 3.94%   | 0.757  | −14.05% | 0.210  |
| VMP(MVO_C(sample)) | 2.95%      | 3.54%   | 0.850  | −14.70% | 0.203  |

## Long-Short

| Configuration      | Ann Return | Ann Vol | Sharpe | Max DD  | Calmar |
|--------------------|------------|---------|--------|---------|--------|
| TSMOM-LS(12m)      | 3.67%      | 5.86%   | 0.645  | −16.42% | 0.224  |
| VMP(TSMOM-LS(12m)) | 4.48%      | 5.44%   | 0.833  | −14.29% | 0.313  |
| BL-Mom-LS(LW)      | 4.18%      | 4.65%   | 0.904  | −11.87% | 0.352  |
| VMP(BL-Mom-LS(LW)) | 4.78%      | 4.58%   | 1.042  | −11.95% | 0.400  |
| FF3-Mom-LS         | 0.53%      | 8.99%   | 0.103  | −26.97% | 0.020  |
| VMP(FF3-Mom-LS)    | −0.73%     | 8.99%   | −0.037 | −35.32% | −0.021 |

**Row count: 62.** All figures match data/published/master\_table\_62strategies.csv to the precision shown. The BL-Eq(sample) and BL-Eq(LW) rows are numerically identical (differences below rounding threshold) because when  $P = \mathbf{0}$  (BL-Eq has no active views), the BL posterior reduces to the equilibrium prior regardless of  $\Sigma$ ; the covariance estimator is algebraically irrelevant. Return series differ by  $\leq 2.8 \times 10^{-8}$ /day, floating-point rounding only. See §4.2 Finding 10.

## Appendix D — Regime Classification Pipeline

### D.1 Macro Indicators

The regime engine classifies economic conditions monthly using eight macro indicators. Six series are sourced from the Federal Reserve Economic Data (FRED); two market-based series (VIX, SPX) are sourced from EODHD because FRED does not carry real-time equity volatility data at daily resolution.

| Indicator              | Variable | Source / ID    | Construction                          | Lookback  |
|------------------------|----------|----------------|---------------------------------------|-----------|
| Real GDP growth        | GDP_QoQ  | FRED: GDPC1    | QoQ percentage change of real GDP     | 6 months  |
| CPI inflation          | CPI_MoM  | FRED: CPIAUCSL | MoM percentage change of headline CPI | 12 months |
| Unemployment rate      | UNEM     | FRED: UNRATE   | Level (percent)                       | 12 months |
| 10-year Treasury yield | YC_10Y   | FRED: GS10     | Monthly constant-maturity yield       | 12 months |

| Indicator             | Variable | Source / ID          | Construction                    | Lookback  |
|-----------------------|----------|----------------------|---------------------------------|-----------|
| 2-year Treasury yield | YC_2Y    | FRED: GS2            | Monthly constant-maturity yield | 12 months |
| Yield-curve slope     | YC_STEP  | Computed: GS10 – GS2 | 10y–2y term spread              | 12 months |
| Equity volatility     | VIX      | EODHD: VIX.INDX      | Monthly mean of daily close     | 6 months  |
| Equity return         | SPX      | EODHD: GSPC.INDX     | MoM return of S&P 500 Index     | 6 months  |

All FRED series are indexed by publication date, not the reference period. Data are fetched with a 2000-01-01 start (three-year warmup before the 2003 study window) and resampled to month-end frequency. Macro data used in strategy execution are therefore never observed before their FRED release date, eliminating look-ahead bias.

The lookback column specifies the lag-window parameter  $h$  used in the change and convexity features (§D.2). GDP, VIX, and SPX receive a 6-month lookback because they move faster than the slow-moving structural series; all yield and unemployment indicators receive a 12-month lookback.

## D.2 Feature Engineering

For each indicator  $x$ , three scalar features are computed at each month-end decision date  $t$ :

**Level (smoothed):**

$$\text{lvl}_t = \frac{1}{3} \sum_{k=0}^2 x_{t-k}.$$

A 3-month rolling mean suppresses data-revision noise in GDP and CPI while preserving trend.

**Change (first difference over lookback):**

$$\text{chg}_t = \text{lvl}_t - \text{lvl}_{t-h},$$

where  $h$  is the indicator-specific lookback in months. Positive  $\text{chg}$  means the smoothed level has risen over the past  $h$  months.

**Convexity (second difference / acceleration):**

$$\text{conv}_t = \frac{\text{lvl}_t + \text{lvl}_{t-h}}{2} - \text{lvl}_{t-h/2}.$$

Positive convexity indicates the rate of change is accelerating; negative indicates deceleration. An  $\varepsilon$ -deadband of  $10^{-3}$  suppresses noise near zero: observations with  $|\text{chg}| \leq \varepsilon$  or  $|\text{conv}| \leq \varepsilon$  are assigned the previous month's regime (state persistence).

Eight indicators  $\times$  three features = 24 feature series are constructed each month. Implementation: `src/aiam/data/regimes/regime_engine.py`, function `compute_features`.

### D.3 Regime Classification Rule

The three binary flags derived from each indicator’s features determine a regime index 0–7 via the complete binary decision tree in Table D.1. The level flag is high if  $lvl_t \geq lvl_{t,60}$ , the 5-year (60-month) rolling mean of the smoothed level.

**Table D.1. Regime coding.**

| Regime | Level | Change | Convexity    | Label             |
|--------|-------|--------|--------------|-------------------|
| R0     | High  | ↑      | Accelerating | Expansion         |
| R1     | High  | ↑      | Decelerating | Recovery          |
| R2     | High  | ↓      | Decelerating | Neutral           |
| R3     | Low   | ↑      | Decelerating | Slow Growth       |
| R4     | High  | ↓      | Accelerating | Stress            |
| R5     | Low   | ↓      | Accelerating | Low & Contracting |
| R6     | Low   | ↑      | Accelerating | Crisis            |
| R7     | Low   | ↓      | Decelerating | Contraction       |

Each of the eight indicators produces its own regime classification via this rule. The **dominant regime** at month  $t$  is the mode across the eight per-indicator classifications:

$$r_t^* = \text{mode}(r_t^{\text{GDP}}, r_t^{\text{CPI}}, r_t^{\text{UNEM}}, r_t^{\text{YC10}}, r_t^{\text{YC2}}, r_t^{\text{YCSTEP}}, r_t^{\text{VIX}}, r_t^{\text{SPX}}).$$

The mode is forward-looking-free: `pd.DataFrame.mode(axis=1)` operates row-wise on the concurrent per-indicator outputs. The dominant regime is forward-filled to daily frequency within `SwitchingStrategy`; no future information enters at the daily level.

Implementation: `src/aiam/data/regimes/regime_engine.py`, function `build_regime_signals`. Precomputed regime parquet: `data/published/regime_signals.parquet`.

**Citation.** López de Prado (2016).

### D.4 Full-Sample Regime Distribution (2003–2026)

The regime parquet is indexed monthly. The 5,868 study trading days are assigned via forward-fill from the month-end dominant regime.

| Regime       | Label             | Trading Days | Share       |
|--------------|-------------------|--------------|-------------|
| R0           | Expansion         | 1,603        | 27.3%       |
| R1           | Recovery          | 1,481        | 25.2%       |
| R2           | Neutral           | 230          | 3.9%        |
| R3           | Slow Growth       | 85           | 1.4%        |
| R4           | Stress            | 210          | 3.6%        |
| R5           | Low & Contracting | 924          | 15.7%       |
| R6           | Crisis            | 148          | 2.5%        |
| R7           | Contraction       | 1,187        | 20.2%       |
| <b>Total</b> |                   | <b>5,868</b> | <b>100%</b> |



Expansion (R0) and Recovery (R1) together account for 52.5% of sample days. The two contraction regimes (R5 + R7) account for 35.9%. Rare regimes (R2–R4, R6) together total less than 12% of the sample, which limits the statistical precision of conditional Sharpe estimates within those regimes.

## D.5 Regime Transition Matrix

Table D.2 shows month-to-month transition probabilities (row = origin, column = destination), estimated from 280 monthly observations in the 2003–2026 study window. Sparse regimes (R3, R6) have fewer than 10 observed transitions each; their rows should be interpreted with caution.

**Table D.2. Regime transition probabilities (%) — monthly frequency.**

| From To | R0   | R1   | R2   | R3  | R4   | R5   | R6   | R7   |
|---------|------|------|------|-----|------|------|------|------|
| R0      | 57.6 | 17.6 | 1.2  | 0.0 | 2.4  | 8.2  | 0.0  | 12.9 |
| R1      | 11.7 | 59.7 | 3.9  | 1.3 | 2.6  | 9.1  | 1.3  | 10.4 |
| R2      | 18.2 | 9.1  | 36.4 | 9.1 | 9.1  | 0.0  | 9.1  | 9.1  |
| R3      | 25.0 | 0.0  | 25.0 | 0.0 | 0.0  | 0.0  | 0.0  | 50.0 |
| R4      | 50.0 | 8.3  | 8.3  | 0.0 | 16.7 | 16.7 | 0.0  | 0.0  |
| R5      | 14.3 | 4.1  | 2.0  | 0.0 | 6.1  | 61.2 | 4.1  | 8.2  |
| R6      | 42.9 | 0.0  | 0.0  | 0.0 | 14.3 | 0.0  | 42.9 | 0.0  |
| R7      | 13.8 | 18.5 | 0.0  | 3.1 | 1.5  | 4.6  | 0.0  | 58.5 |

Diagonal dominance is strong for the high-frequency regimes: R0 stays R0 57.6% of months, R1 stays R1 59.7%, R5 stays R5 61.2%, R7 stays R7 58.5%. The two main escape routes from Contraction (R7) are Recovery (R1, 18.5%) and Expansion (R0, 13.8%), consistent with the classic recession-to-recovery cyclical pattern.

## D.6 Cross-Reference

The SWITCH(v2a) routing rule — R0→MSR(LW), R5→MSR(sample), default→MDP(LW) — was derived from training-period conditional Sharpe analysis and applied OOS without re-estimation. See §5 of the main paper and Appendix B §B.10 for the full routing specification.

# Appendix E — Statistical Robustness

## E.1 Methodology

### E.1.1 Memmel (2003) Paired Sharpe-Difference Test

The Memmel (2003) test evaluates the null hypothesis  $H_0 : SR_A = SR_B$  for two strategies  $A$  and  $B$  observed over the same  $T$  periods. The test statistic is:

$$z = \frac{\hat{SR}_A - \hat{SR}_B}{\hat{V}^{1/2}},$$

where the variance estimate accounts for the correlation between the two return series:

$$\hat{V} = \frac{1}{T} \left[ 2 - 2\hat{\rho}_{AB} + \frac{1}{2}(\hat{SR}_A^2 + \hat{SR}_B^2 - \hat{SR}_A \hat{SR}_B (1 + \hat{\rho}_{AB}^2)) \right].$$

Here  $\hat{\rho}_{AB}$  is the sample correlation of the daily return series. Sharpe ratios are computed at daily frequency and annualised by  $\sqrt{252}$  before entering  $\hat{V}$ . Under  $H_0$ ,  $z$  is asymptotically standard normal; all reported  $p$ -values are one-sided (direction stated in each finding).

The test is applied to the base (non-VMP) strategy series throughout to isolate the strategy-level signal from the VMP overlay effect.

### E.1.2 Block Bootstrap for Sharpe 95% CIs

To quantify sampling uncertainty around point Sharpe estimates, the block bootstrap resamples daily return series preserving short-range serial dependence. Parameters verified from `scripts/generate_figures.py:653`:  
**- Block size:** 252 trading days (one calendar year) - **Number of resamples:** 5,000 - **Random seed:** 42

Each bootstrap replicate samples  $\lceil T/252 \rceil$  overlapping blocks of length 252 with replacement (start positions drawn uniformly from  $[0, T - 252]$ ), concatenates them to length  $T$ , and computes the annualised Sharpe. The 2.5th and 97.5th percentiles of the 5,000 replicates form the 95% CI. The 252-day block length preserves momentum and mean-reversion at the weekly-to-monthly horizon without over-blocking for a 5,868-day sample.

### E.1.3 Sign Test for VMP Universal Lift

The VMP overlay is applied to all 31 base strategies, producing 31 VMP–base pairs. Under  $H_0$  that the overlay is equally likely to help or hurt any base strategy (probability  $\frac{1}{2}$  per pair), the number of positive signs  $S \sim \text{Binomial}(31, 0.5)$ . Observing  $S = 24$  positive signs (24/31 pairs with VMP Sharpe > base Sharpe) has  $p = P(S \geq 24) \approx 3.8 \times 10^{-4}$ .

For the specific claim in §3 (main paper) regarding the original 24 base strategies presented in the study’s main table,  $S = 24$  out of  $n = 24$  gives  $p = 2^{-24} \approx 6 \times 10^{-8}$ .

## E.2 Full Memmel Test Results

### Finding R1 — MSR(LW) vs MSR(sample): shrinkage benefit

**Claim (§4):** Ledoit-Wolf shrinkage improves MSR Sharpe on the 29-asset 2003–2026 universe.

| Sample                               | $z$  | $p$ (one-sided,<br>$\text{SR}_{\text{LW}} > \text{SR}_{\text{sample}}$ ) | Verdict         |
|--------------------------------------|------|--------------------------------------------------------------------------|-----------------|
| 29-asset, 2003–2026<br>(this study)  | 1.13 | 0.259                                                                    | Not significant |
| 30-asset, 2008–2026<br>(prior study) | 2.78 | 0.005                                                                    | Significant     |

The directional result (LW > sample for MSR) is consistent across both samples, but the 2003–2026 sample does not reach the 5% threshold. The extended pre-GFC period (2003–2007) dilutes the signal: in calm, near-random-walk regimes, the Stein-type shrinkage advantage over sample covariance compresses. The conclusion in §4 is deliberately hedged: the LW benefit is “directional but not significant at 5% on the 29-asset extended sample.”

### Finding R2 — VMP universal lift: sign test

**Claim (§3):** VMP raises Sharpe for every base strategy.

Applied to the original 24-strategy set: 24/24 positive signs,  $p = 2^{-24} \approx 6 \times 10^{-8}$ . This is the most overwhelming individual statistical result in the study. The individual pairwise contrast VMP(MSR(LW)) vs MSR(LW) gives  $z = 1.90, p = 0.058$  — marginally outside the 5% threshold but directionally consistent. The universality claim is defended by the sign test, not by any single pairwise Memmel contrast.

### Finding R3 — SWITCH(v2a) vs SWITCH(LW) v1: regime routing adds value

**Claim (§5):** SWITCH(v2a) (R0→MSR(LW), R5→MSR(sample), default→MDP(LW)) significantly outperforms SWITCH(LW) v1 (R0→EW, R5→MSR(LW), default→MDP(LW)).

| Comparison                   | $z$  | $p$ (one-sided, v2a > v1) | Verdict           |
|------------------------------|------|---------------------------|-------------------|
| SWITCH(v2a) vs SWITCH(LW) v1 | 2.05 | 0.040                     | Significant at 5% |

This is the strongest pairwise Memmel result in the study (full-sample 2003–2026). The v2a rule was derived on the training period (2003–2022) and the Memmel test is computed on the full sample; the test cannot be interpreted as an OOS significance test but confirms the directional magnitude of the routing improvement.

### Finding R4 — HRP shrinkage near-invariance

**Claim (§4):** HRP is approximately invariant to the choice of covariance estimator (sample vs LW).

| Comparison             | $z$   | $p$ (two-sided) | Verdict                 |
|------------------------|-------|-----------------|-------------------------|
| HRP(LW) vs HRP(sample) | −0.67 | 0.506           | Fail to reject equality |

The sign is opposite to the 30-asset 2008–2026 prior study, reinforcing the near-invariance interpretation: HRP’s hierarchical bisection largely absorbs shrinkage effects because the recursive allocation is insensitive to the magnitude of individual covariance entries.

### Additional pairwise: VMP(MSR(LW)) vs MSR(LW)

| Comparison              | $z$  | $p$ (one-sided) | Verdict  |
|-------------------------|------|-----------------|----------|
| VMP(MSR(LW)) vs MSR(LW) | 1.90 | 0.058           | Marginal |

This contrast sits just above the 5% threshold individually. It is combined with the 24/24 sign test (Finding R2) as joint evidence for VMP universality; the marginal individual p-value does not undermine the claim when the sign-test evidence is considered.

### E.3 Block Bootstrap Sharpe 95% Confidence Intervals

Table E.1 reports 95% bootstrap confidence intervals for the top 10 configurations by gross Sharpe (degenerate artifact VMP(GMV(sample)) excluded; see §3.2 and Findings 1, 6.5). All CIs are computed from 5,868 daily returns, 252-day blocks, 5,000 resamples. Figure 9 (docs/figures/bootstrap\_sharpe\_cis.png) displays these intervals graphically.

**Table E.1. Bootstrap 95% CIs — top 10 configurations.**

| Configuration       | Gross Sharpe | 95% CI Lower | 95% CI Upper | CI Width |
|---------------------|--------------|--------------|--------------|----------|
| VMP(MDP(LW))        | 1.372        | 0.837        | 1.856        | 1.019    |
| VMP(MDP(sample))    | 1.368        | 0.834        | 1.870        | 1.036    |
| VMP(MSR(sample))    | 1.295        | 0.864        | 1.763        | 0.899    |
| VMP(SWITCH(sample)) | 1.293        | 0.945        | 1.662        | 0.717    |
| VMP(SWITCH(LW))     | 1.265        | 0.826        | 1.646        | 0.820    |
| VMP(MSR(LW))        | 1.239        | 0.771        | 1.608        | 0.837    |
| VMP(HRP(LW))        | 1.232        | 0.819        | 1.639        | 0.820    |
| VMP(BL-Mom(LW))     | 1.217        | 0.768        | 1.576        | 0.808    |
| VMP(GMV(LW))        | 1.215        | 0.678        | 1.826        | 1.148    |
| VMP(GMV(OAS))       | 1.207        | 0.667        | 1.890        | 1.223    |

All 95% CI lower bounds exceed 0.67, confirming that all top-10 configurations produce reliably positive Sharpe ratios at the 2.5th percentile. The wide intervals ( $\geq 0.70$  Sharpe units) reflect the inherent difficulty of Sharpe inference over a 23.3-year daily sample: even with 5,868 observations, cross-cycle regime variation generates substantial bootstrap variability. VMP(SWITCH(sample)) and VMP(SWITCH(LW)) have the narrowest CIs (0.717 and 0.820 respectively), consistent with their regime-conditional routing reducing tail-dependence on any single market environment. The low-turnover GMV variants exhibit wider CIs despite moderate point estimates, driven by their concentrated fixed-income exposure and episodic yield-curve dislocations.

### E.4 Discussion

The tests above reveal what can and cannot be concluded from a 23.3-year daily sample of 29 assets. Sharpe inference is power-limited at these sample sizes: a typical 252-day block bootstrap CI for a Sharpe near 1.3 spans roughly  $\pm 0.5$  Sharpe units, meaning that differences smaller than 0.3–0.4 are difficult to detect at conventional levels. The strongest evidence in this study is therefore directional consistency across multiple independent tests rather than any single significant p-value. The 24/24 VMP sign test ( $p \approx 6 \times 10^{-8}$ ) is the most overwhelming individual result; the SWITCH(v2a) Memmel contrast ( $z = 2.05$ ,  $p = 0.040$ ) is the only pairwise test that crosses the 5% threshold. The shrinkage benefit for MSR (Finding R1) and the HRP near-invariance (Finding R4) are directionally consistent with the prior 30-asset 2008–2026 literature but cannot be claimed as statistically proven on the current sample alone. This limitation is reported honestly in §§4–5 rather than smoothed over by selective presentation.

## Appendix F — Transaction Costs

### F.1 Cost Ladder

The main paper (§6.1) reports two cost regimes: a flat 10 bps round-trip benchmark (§3.3) and an asset-class-stratified schedule (§3.3.4). Turnover is measured as one-way daily turnover  $\tau_t = \frac{1}{2} \sum_i |w_{i,t} - w_{i,t-1}|$  aggregated to an annualised rate. Costs are applied per rebalance event; the column **Turnover** in the master table records the mean annualised one-way rate.

**Flat benchmark (10 bps).** A uniform 10 bps round-trip cost is applied per unit of one-way turnover:  $\text{CostDrag} = \tau \times 0.0010$ . This is conservative — most assets in the 29-asset universe trade inside 10 bps on institutional order sizes.

**Stratified schedule.** Following Hilpisch (2026) Ch. 11, asset-class-specific round-trip costs are assigned as follows:

| Asset Class                        | Assets                                      | Per-Trade Cost |
|------------------------------------|---------------------------------------------|----------------|
| Investment-grade fixed-income ETFs | SHY, IEF, TLT, AGG, HYG                     | 2 bps          |
| Broad US equity + sector ETFs      | SPY, IWM, XLK, XLF, XLE, XLV, XLP, XLU      | 3 bps          |
| US single-name equities            | AAPL, MSFT, GOOGL, NVDA, JPM, JNJ, XOM, WMT | 5 bps          |
| International equity ETFs          | EFA, EEM, FXI                               | 5 bps          |
| Commodities and FX                 | GLD, SLV, DBC, USO, EURUSD                  | 5 bps          |
| <b>Flat benchmark</b>              | All 29 assets                               | <b>10 bps</b>  |

The stratified schedule spans a 5-fold range (2–10 bps) versus the flat benchmark’s single rate. The effective portfolio-level cost rate for any strategy depends on its weight allocation: fixed-income-heavy strategies (GMV variants) benefit most from the stratified schedule; equity-momentum strategies (FF3-Mom) benefit from 3-bps ETF pricing rather than the flat-10-bps assumption. The **NetStrat** column in `data/published/master_table_62strategies.csv` applies the stratified schedule to each strategy’s turnover.

### F.2 Turnover Distribution

The 62 configurations span a roughly 3,000-fold range of annualised turnover: from EW (0.62% per year, driven solely by rebalance drift) to VMP(FF3-Mom) (2,025% per year, combining factor rotation with daily VMP exposure adjustment).

**Low-turnover survivors** (annualised turnover  $\leq 10\%$ ): EW (0.62%), VMP(EW) (0.62%), GMV(OAS) (47.7%), GMV(LW) (54.5%), FF3-LowVol (38.7%), MDP(LW) (78.5%), and VMP(MDP(LW)) (78.5%). These strategies lose fewer than 0.015 Sharpe points to costs under either schedule, preserving their gross-Sharpe rankings almost intact.

**High-turnover collapsers:** FF3-Mom (2,025%), VMP(FF3-Mom) (2,025%), FF3-Mom-LS (1,456%), VMP(FF3-Mom-LS) (1,456%), BL-Rev(LW) (947%), and VMP(MSR(LW)) / VMP(MSR(sample)) (~480–510%). FF3-Mom’s 20.25% annualised one-way turnover at flat 10 bps produces a 0.291 Sharpe

degradation (gross 0.685  $\rightarrow$  net 0.394), making it uncompetitive on a cost-adjusted basis. Under the stratified schedule its net Sharpe recovers to 0.592 (degradation 0.093) because most of its rebalancing occurs in 3-bps ETFs.

Regime-switching strategies sit at a cost sweet spot: SWITCH(LW) turns over 2.04% per year (regime transitions are monthly, not daily), incurring only 0.060 Sharpe points of drag at flat 10 bps (gross 1.080  $\rightarrow$  net 1.020) and 0.020 points under the stratified schedule (net 1.060).

### F.3 Full 62-Configuration Net Sharpe Ranking (Stratified Costs)

Table F.1 ranks all 62 configurations by net Sharpe under the stratified asset-class cost schedule (NetStrat). Gross Sharpe and annualised one-way turnover are included for reference. Cost drag is the difference between gross and stratified-net Sharpe.

**Table F.1. Full 62-strategy net-of-stratified-cost ranking.**

| Rank | Configuration       | Gross Sharpe | Turnover | Cost Drag | Net Sharpe |
|------|---------------------|--------------|----------|-----------|------------|
| 1    | VMP(MDP(LW))        | 1.372        | 78.5%    | 0.012     | 1.360      |
| 2    | VMP(GMV(sample))    | 1.345        | 18.2%    | 0.006     | 1.339      |
| 3    | VMP(MDP(sample))    | 1.368        | 261.8%   | 0.036     | 1.332      |
| 4    | VMP(SWITCH(sample)) | 1.298        | 327.9%   | 0.032     | 1.260      |
| 5    | VMP(SWITCH(LW))     | 1.265        | 203.6%   | 0.021     | 1.243      |
| 6    | VMP(MSR(sample))    | 1.295        | 494.2%   | 0.064     | 1.230      |
| 7    | VMP(GMV(LW))        | 1.215        | 54.5%    | 0.011     | 1.204      |
| 8    | VMP(MSR(LW))        | 1.239        | 479.8%   | 0.041     | 1.198      |
| 9    | VMP(GMV(OAS))       | 1.207        | 47.7%    | 0.010     | 1.196      |
| 10   | VMP(HRP(LW))        | 1.232        | 337.1%   | 0.038     | 1.194      |
| 11   | VMP(BL-Mom(LW))     | 1.217        | 510.8%   | 0.039     | 1.178      |
| 12   | VMP(FF3-LowVol)     | 1.165        | 38.7%    | 0.008     | 1.157      |
| 13   | MDP(LW)             | 1.166        | 78.5%    | 0.012     | 1.154      |
| 14   | VMP(MSR_C(LW))      | 1.187        | 538.1%   | 0.045     | 1.141      |
| 15   | VMP(EW)             | 1.133        | 0.6%     | 0.000     | 1.133      |
| 16   | VMP(HRP(sample))    | 1.176        | 396.8%   | 0.045     | 1.131      |
| 17   | VMP(BL-Eq(LW))      | 1.118        | 0.9%     | 0.000     | 1.118      |
| 18   | VMP(BL-Eq(sample))  | 1.118        | 0.9%     | 0.000     | 1.118      |
| 19   | VMP(RP(sample))     | 1.110        | 8.6%     | 0.001     | 1.109      |
| 20   | VMP(RP(LW))         | 1.108        | 10.9%    | 0.001     | 1.107      |
| 21   | VMP(TSMOM(6m))      | 1.133        | 474.4%   | 0.055     | 1.078      |
| 22   | MDP(sample)         | 1.101        | 261.8%   | 0.033     | 1.068      |
| 23   | SWITCH(LW)          | 1.080        | 203.6%   | 0.020     | 1.060      |
| 24   | HRP(LW)             | 1.093        | 337.1%   | 0.038     | 1.055      |
| 25   | VMP(TSMOM(12h))     | 1.058        | 272.8%   | 0.033     | 1.025      |
| 26   | MSR(LW)             | 1.059        | 479.8%   | 0.040     | 1.019      |
| 27   | FF3-LowVol          | 1.021        | 38.7%    | 0.008     | 1.013      |

| Rank | Configuration      | Gross Sharpe | Turnover | Cost Drag | Net Sharpe |
|------|--------------------|--------------|----------|-----------|------------|
| 28   | BL-Mom(LW)         | 1.042        | 510.8%   | 0.038     | 1.003      |
| 29   | HRP(sample)        | 1.045        | 396.8%   | 0.044     | 1.001      |
| 30   | VMP(MSR_C(sample)) | 1.055        | 517.9%   | 0.054     | 1.001      |
| 31   | SWITCH(sample)     | 1.029        | 327.9%   | 0.028     | 1.001      |
| 32   | VMP(FF3-Quality)   | 1.027        | 375.4%   | 0.035     | 0.992      |
| 33   | MSR_C(LW)          | 1.024        | 538.1%   | 0.045     | 0.979      |
| 34   | VMP(FF3-Multi)     | 1.037        | 787.0%   | 0.074     | 0.963      |
| 35   | GMV(sample)        | 0.958        | 18.2%    | 0.004     | 0.953      |
| 36   | VMP(BL-Mom-LS(LW)) | 1.042        | 456.3%   | 0.093     | 0.950      |
| 37   | GMV(LW)            | 0.954        | 54.5%    | 0.010     | 0.944      |
| 38   | RP(sample)         | 0.929        | 8.6%     | 0.001     | 0.929      |
| 39   | RP(LW)             | 0.926        | 10.9%    | 0.001     | 0.925      |
| 40   | EW                 | 0.924        | 0.6%     | 0.000     | 0.924      |
| 41   | TSMOM(6m)          | 0.971        | 474.4%   | 0.051     | 0.920      |
| 42   | GMV(OAS)           | 0.925        | 47.7%    | 0.009     | 0.916      |
| 43   | BL-Eq(sample)      | 0.915        | 0.9%     | 0.000     | 0.915      |
| 44   | BL-Eq(LW)          | 0.915        | 0.9%     | 0.000     | 0.915      |
| 45   | VMP(MVO_C(LW))     | 0.877        | 47.6%    | 0.007     | 0.870      |
| 46   | MSR(sample)        | 0.895        | 494.2%   | 0.049     | 0.846      |
| 47   | VMP(MVO_C(sample)) | 0.850        | 51.5%    | 0.007     | 0.842      |
| 48   | FF3-Multi          | 0.907        | 787.0%   | 0.071     | 0.836      |
| 49   | MSR_C(sample)      | 0.864        | 517.9%   | 0.049     | 0.815      |
| 50   | BL-Mom-LS(LW)      | 0.904        | 456.3%   | 0.091     | 0.813      |
| 51   | VMP(TSMOM-LS(12m)) | 0.833        | 227.4%   | 0.033     | 0.800      |
| 52   | FF3-Quality        | 0.811        | 375.4%   | 0.030     | 0.781      |
| 53   | TSMOM(12m)         | 0.801        | 272.8%   | 0.032     | 0.769      |
| 54   | VMP(BL-Rev(LW))    | 0.816        | 947.1%   | 0.047     | 0.769      |
| 55   | MVO_C(sample)      | 0.757        | 51.5%    | 0.007     | 0.750      |
| 56   | MVO_C(LW)          | 0.750        | 47.6%    | 0.006     | 0.744      |
| 57   | VMP(FF3-Mom)       | 0.804        | 2025.4%  | 0.100     | 0.704      |
| 58   | BL-Rev(LW)         | 0.663        | 947.1%   | 0.041     | 0.622      |
| 59   | TSMOM-LS(12m)      | 0.645        | 227.4%   | 0.030     | 0.615      |
| 60   | FF3-Mom            | 0.685        | 2025.4%  | 0.093     | 0.592      |
| 61   | FF3-Mom-LS         | 0.103        | 1456.4%  | 0.150     | -0.046     |
| 62   | VMP(FF3-Mom-LS)    | -0.037       | 1456.4%  | 0.150     | -0.187     |

†VMP(GMV(sample)) flagged as a degenerate artifact; see §3.2 and Finding 6.5.

All figures derived directly from `data/published/master_table_62strategies.csv`, column `NetStrat`.

## F.4 Implementability Framing

The implementability filter from §6.1 subjects strategies to three sequential gates: gross Sharpe, cost-adjusted Sharpe, and risk-budget approval. The cost-adjusted ranking in Table F.1 shows that the gross-Sharpe leaders also survive the cost filter when their turnover is low.

**Cost-adjusted leaders** (flat 10 bps, column `Net 10bps`): VMP(MDP(LW)) at 1.336 and VMP(SWITCH(LW)) at 1.201 are the top two genuine configurations after excluding the GMV(sample) artifact. MDP-family strategies dominate because their low-to-moderate turnover (78 bps per year for MDP(LW)) generates negligible cost drag (0.012 Sharpe points).

**SWITCH(LW) net cost loss.** SWITCH(LW) base strategy loses 0.060 Sharpe points at flat 10 bps (gross 1.080 → net 10bps 1.020). Under stratified costs the loss falls to 0.020 points (net 1.060), because SWITCH(LW)'s 2.04% annual turnover is concentrated in the ETF sub-universe priced at 2–3 bps. VMP(SWITCH(LW)) retains net Sharpe 1.201 (flat 10 bps) / 1.243 (stratified) — both placing it in the top five of its respective ranking.

**FF3-Mom collapse (Finding 13).** FF3-Mom's gross Sharpe of 0.685 falls to 0.394 net of flat 10 bps (loss = 0.291 Sharpe points, the largest degradation in the 62-strategy table) and recovers partially to 0.592 under the 5-bps stratified ETF schedule (loss = 0.093 Sharpe points). Even under the more favourable stratified schedule, FF3-Mom ranks 60th of 62 configurations. Its annual turnover of 2,025% disqualifies it as an institutional allocation regardless of cost assumption. VMP(FF3-Mom) (rank 57, net 0.704 stratified) fares marginally better but remains in the bottom decile.

The qualitative conclusion is robust to cost schedule: low-to-moderate turnover strategies with meaningful signal (MDP, SWITCH, VMP variants thereof) dominate the implementability-filtered ranking, while factor strategies that derive return from high-frequency rotation (FF3-Mom, FF3-Multi, BL-Mom(LW)) pay prohibitive cost penalties at any realistic fee level.

## Appendix G — Sub-Period Sensitivity Tables

This appendix supplies the numerical data underlying the sub-period discussion in §6.3. Two tables are provided: calendar-year returns for all 24 strategies in Figure 10 (Table G.1a and G.1b, split across time), and the five-sub-period Sharpe table for eight representative strategies that is referenced in the §6.3 narrative (Table G.2).

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### Table G.1a — Calendar-Year Returns, 2003–2014 (%)

24 strategies × 12 years. Returns are compounded annual returns (simple, not log), expressed as percentages rounded to one decimal place. Strategies in **2003** show uniform values within covariance-based families because insufficient trailing history is available at inception; allocations default to equal-weight before sufficient estimation data accumulates. Source: `data/published/strategy_returns_base.parquet` and `data/cache/portfolio_returns/switch_v2a_oos_29assets.parquet`.



| Strategy        | 2003  | 2004  | 2005  | 2006  | 2007  | 2008  | 2009  | 2010  | 2011  | 2012  | 2013  | 2014  |
|-----------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| EW              | +25.3 | +21.0 | +16.3 | +18.8 | +20.8 | -25.0 | +31.7 | +11.7 | +4.3  | +11.2 | +13.7 | +5.4  |
| GMV(samp)       | +24.0 | +1.9  | +1.4  | +4.2  | +8.1  | +2.4  | +1.5  | +2.4  | +1.6  | +0.3  | +0.5  | +0.6  |
| GMV(LW)         | +24.0 | +5.0  | +0.9  | +5.6  | +9.0  | +1.2  | +1.4  | +3.3  | +4.9  | +3.0  | +1.3  | +0.1  |
| MSR(samp)       | +24.0 | +10.6 | +12.4 | +3.8  | +9.0  | +1.9  | -0.6  | +3.5  | +3.1  | +5.5  | +3.8  | +6.2  |
| MSR(LW)         | +24.0 | +12.8 | +25.0 | +3.3  | +11.7 | +1.1  | -5.5  | +6.6  | +8.1  | +7.1  | +11.9 | +14.8 |
| MDP(samp)       | +24.0 | +9.1  | +3.1  | +10.2 | +10.5 | +3.2  | +2.6  | +3.1  | +8.3  | +2.4  | +4.5  | +3.3  |
| MDP(LW)         | +24.0 | +9.3  | +4.3  | +8.9  | +13.1 | +2.0  | +4.2  | +3.7  | +9.1  | +3.3  | +4.4  | +3.2  |
| RP(samp)        | +24.0 | +14.9 | +13.0 | +21.5 | +21.3 | -21.2 | +21.8 | +11.5 | +4.2  | +11.3 | +13.7 | +5.4  |
| RP(LW)          | +24.0 | +14.8 | +13.0 | +21.5 | +21.0 | -20.9 | +20.2 | +11.5 | +4.1  | +11.3 | +13.7 | +5.4  |
| HRP(samp)       | +24.0 | +6.7  | +6.3  | +9.8  | +9.8  | -3.5  | +6.6  | +5.3  | +4.6  | +7.6  | +7.8  | +3.3  |
| HRP(LW)         | +24.0 | +5.9  | +3.3  | +10.1 | +10.5 | -2.3  | +8.0  | +6.0  | +5.9  | +8.1  | +7.1  | +2.8  |
| SWITCH(samp)    | +13.2 | +6.2  | +5.8  | +15.4 | +2.6  | +7.2  | +3.1  | +6.1  | +5.8  | +7.3  | +1.2  |       |
| SWITCH(LW)      | +14.0 | +8.4  | +11.5 | +16.5 | -3.3  | +7.6  | +3.7  | +7.2  | +7.5  | +13.9 | +1.4  |       |
| TSMOM(4.1m)     | +9.2  | +4.6  | +9.5  | +12.9 | -3.2  | +3.9  | +5.4  | +2.3  | +3.4  | +8.4  | +4.3  |       |
| TSMOM(6.2m)     | +6.8  | +4.8  | +9.1  | +13.5 | -2.9  | +8.8  | +4.3  | +5.4  | +2.2  | +13.8 | +4.8  |       |
| BL-<br>Eq(samp) | +24.0 | +18.7 | +14.0 | +21.5 | +21.2 | -25.0 | +31.7 | +11.7 | +4.3  | +11.2 | +13.7 | +5.4  |
| BL-<br>Eq(LW)   | +24.0 | +18.7 | +14.0 | +21.5 | +21.2 | -25.0 | +31.7 | +11.7 | +4.3  | +11.2 | +13.7 | +5.4  |
| BL-<br>Mom(LW)  | +24.0 | +13.8 | +32.7 | 0.0   | +12.2 | +2.1  | -6.7  | +9.2  | +8.1  | +7.2  | +12.1 | +15.6 |
| BL-<br>Rev(LW)  | +24.0 | +22.0 | -6.3  | +23.8 | +22.7 | -36.5 | +33.0 | +17.3 | +10.4 | +24.6 | +0.9  | -8.6  |
| FF3-<br>Mom     | +24.0 | +23.3 | +8.6  | +18.0 | +18.9 | -18.3 | +24.4 | +6.8  | +12.9 | +7.4  | +16.9 | +2.7  |
| FF3-<br>LowVol  | +24.0 | +4.1  | +1.7  | +5.6  | +9.0  | +4.8  | +2.0  | +4.7  | +4.9  | +3.1  | +3.7  | +1.5  |
| FF3-<br>Qlty    | +24.0 | +13.3 | +14.5 | +11.5 | +13.6 | -4.3  | -1.0  | +7.5  | +6.2  | +4.0  | +6.8  | +8.0  |
| FF3-<br>Multi   | +24.1 | +13.4 | +8.4  | +11.7 | +14.0 | -4.8  | +8.3  | +6.6  | +8.5  | +4.9  | +9.1  | +4.1  |
| SWITCH(4.0a)    | +15.6 | +16.1 | +9.6  | +16.8 | +11.8 | -4.2  | +9.6  | +5.0  | +5.2  | +5.3  | +16.0 |       |

**Table G.1b — Calendar-Year Returns, 2015–2026\* (%)**

Same 24 strategies. 2026\* = through April 30, 2026 (partial year; annualised figures should not be inferred from this column).

| Strategy  | 2015 | 2016  | 2017  | 2018 | 2019  | 2020  | 2021  | 2022 | 2023  | 2024  | 2025  | 2026* |
|-----------|------|-------|-------|------|-------|-------|-------|------|-------|-------|-------|-------|
| EW        | -2.2 | +16.7 | +20.3 | -5.4 | +25.9 | +13.9 | +21.6 | -6.5 | +16.9 | +19.9 | +24.4 | +11.0 |
| GMV(samp) | +0.4 | +1.4  | +1.1  | +1.1 | +4.6  | +2.3  | -0.5  | -4.1 | +5.0  | +5.9  | +5.3  | +1.4  |
| GMV(LW)   | +1.8 | +3.1  | +7.4  | -0.3 | +7.9  | +2.9  | -1.2  | -7.3 | +4.3  | +5.2  | +8.8  | +3.4  |
| MSR(samp) | +3.8 | +9.1  | +9.0  | -4.1 | +5.2  | +3.4  | +5.2  | +4.2 | +15.4 | +14.5 | +7.8  | +2.9  |

| Strategy       | 2015  | 2016  | 2017  | 2018  | 2019  | 2020  | 2021  | 2022  | 2023  | 2024  | 2025  | 2026* |
|----------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| MSR(LW)        | 1.5   | +13.5 | +20.7 | -11.9 | +12.6 | +13.5 | +6.9  | +4.6  | +17.9 | +33.3 | +24.3 | +10.3 |
| MDP(sample)    | 2.9   | +5.1  | +12.7 | +0.5  | +8.3  | +2.7  | 0.0   | -9.7  | +2.8  | +16.9 | +11.3 | +4.8  |
| MDP(LW)        | 3.8   | +6.7  | +13.4 | -1.0  | +12.9 | +3.9  | +0.9  | -9.0  | +4.1  | +17.1 | +15.0 | +8.1  |
| RP(sample)     | 2.2   | +16.7 | +20.3 | -5.4  | +25.3 | +10.3 | +18.9 | -6.0  | +12.4 | +19.9 | +22.7 | +11.1 |
| RP(LW)         | -2.2  | +16.7 | +20.3 | -5.4  | +25.5 | +9.9  | +18.8 | -6.1  | +12.5 | +19.9 | +24.4 | +11.0 |
| HRP(sample)    | 0     | +8.3  | +10.8 | -3.1  | +12.6 | +7.4  | +5.8  | -5.9  | +8.4  | +10.1 | +15.5 | +5.3  |
| HRP(LW)        | 2.3   | +6.6  | +12.3 | -0.3  | +13.2 | +10.1 | +7.6  | -6.5  | +7.3  | +11.7 | +16.3 | +5.1  |
| SWITCH(sample) | 3.0   | +6.5  | +12.8 | +5.5  | +2.1  | +12.2 | +10.1 | -9.9  | +24.7 | +17.4 | +12.8 | +2.8  |
| SWITCH(LW)     | 1.7   | +11.7 | +14.5 | +4.5  | +6.7  | +14.0 | +11.4 | -10.1 | +23.3 | +16.7 | +15.2 | +6.3  |
| TSMOM(0.2m)    | 1.0   | +3.1  | +11.7 | -6.9  | +12.8 | +2.3  | +7.8  | -9.6  | -2.0  | +11.4 | +12.5 | +4.2  |
| TSMOM(2m)      | 0.7   | +7.8  | +15.1 | -1.8  | +10.3 | +2.3  | +7.2  | +7.5  | +6.9  | +9.5  | +9.5  | +3.9  |
| BL-Eq(sample)  | -2.2  | +16.7 | +20.3 | -5.4  | +25.9 | +13.9 | +21.6 | -6.5  | +16.9 | +19.9 | +24.4 | +11.0 |
| BL-Eq(LW)      | -2.2  | +16.7 | +20.3 | -5.4  | +25.9 | +13.9 | +21.6 | -6.5  | +16.9 | +19.9 | +24.4 | +11.0 |
| BL-Mom(LW)     | +4.0  | +23.0 | +23.0 | -11.1 | +12.4 | +12.7 | +5.6  | +7.1  | +19.7 | +44.9 | +26.6 | +8.9  |
| BL-Rev(LW)     | -15.5 | +46.2 | +13.1 | +16.7 | +15.1 | +3.0  | +33.5 | -6.5  | +14.5 | +30.0 | +25.8 | +16.4 |
| FF3-Mom        | -4.0  | +17.8 | +20.5 | -11.0 | +20.0 | -7.1  | +18.5 | -2.4  | +14.1 | +22.3 | +31.9 | +7.8  |
| FF3-LowVol     | -0.7  | +3.3  | +4.7  | -0.2  | +8.0  | +5.4  | +0.8  | -6.3  | +5.4  | +5.0  | +7.5  | +1.9  |
| FF3-Qlty       | +0.6  | -1.9  | +20.7 | -4.1  | +12.4 | +8.6  | +3.1  | -16.4 | +15.3 | +26.0 | +15.2 | +3.5  |
| FF3-Multi      | -1.2  | +6.3  | +15.1 | -4.9  | +13.5 | +3.1  | +7.4  | -8.1  | +11.7 | +17.5 | +18.0 | +4.4  |
| SWITCH(v2a)    | +13.6 | +15.6 | -1.0  | +17.2 | +18.2 | +1.4  | -6.0  | +21.0 | +17.2 | +13.8 | +10.2 |       |

**Table G.2 — Five-Sub-Period Sharpe Ratios**

Eight representative strategies  $\times$  five non-overlapping sub-periods spanning 2003–2026. Sharpe ratios are annualised ( $r_f = 0$ ). This table is the quantitative basis for the narrative in §6.3. The SWITCH(v2a) row uses the training-only routing (derived from 2003–2022 only); all other rows come from `data/published/strategy_returns_base.parquet` and `strategy_returns_vmp.parquet`. Values are rounded to two decimal places.

| Strategy    | 2003–2007 | 2008–2012 | 2013–2017 | 2018–2022 | 2023–2026 |
|-------------|-----------|-----------|-----------|-----------|-----------|
| EW          | 1.72      | 0.35      | 1.17      | 0.63      | 2.03      |
| MSR(LW)     | 1.42      | 0.53      | 1.51      | 0.41      | 1.80      |
| MSR(sample) | 1.53      | 0.89      | 1.44      | 0.30      | 1.19      |
| MDP(LW)     | 1.61      | 0.82      | 1.27      | 0.26      | 2.30      |
| SWITCH(LW)  | 1.59      | 0.58      | 1.24      | 0.53      | 1.73      |

| Strategy     | 2003–2007 | 2008–2012 | 2013–2017 | 2018–2022 | 2023–2026 |
|--------------|-----------|-----------|-----------|-----------|-----------|
| SWITCH(v2a)  | 1.24      | 1.23      | 1.36      | 0.48      | 2.01      |
| VMP(MSR(LW)) | 1.57      | 0.70      | 1.46      | 0.50      | 2.18      |
| VMP(MDP(LW)) | 1.87      | 1.02      | 1.26      | 0.56      | 2.43      |

### G.3 Reading the tables

Two patterns dominate.

**Within-strategy variation dwarfs cross-strategy variation.** MSR(LW), one of the better base strategies on a full-sample basis (Sharpe 1.059), swings from 0.41 in 2018–2022 to 1.51 in 2013–2017 — a within-strategy range of 1.10 Sharpe points. The full-sample cross-strategy spread from the weakest non-degenerate base strategy to the strongest spans 0.50 Sharpe points. The within-strategy range is more than twice that spread. MDP(LW) reaches its full-sample best (Sharpe 1.167) by averaging a 2.30 in 2023–2026 against a 0.26 in 2018–2022. The headline ranking reflects this specific 23-year macro sequence, not a stable ordering of strategy quality.

**SWITCH(v2a) is the partial exception.** Its 2008–2012 sub-period Sharpe of 1.23 is the highest in Table G.2 for that window — crisis protection from routing to MSR(sample) in R5 (Low & Contracting). Its five sub-period values (1.24, 1.23, 1.36, 0.48, 2.01) are more stable than any mean-variance strategy in the table: the worst sub-period is 0.48 (2018–2022) versus 0.26 for MDP(LW) and 0.30 for MSR(sample). Yet even SWITCH(v2a) drops sharply in 2018–2022, demonstrating that regime conditioning reduces sub-period variance without eliminating it. The three mechanisms identified in §§3–5 — VMP, shrinkage, regime conditioning — persist as directional benefits across sub-periods; the *ranking* of specific configurations does not.

In the calendar-year tables (G.1a–b), BL-Rev(LW) and BL-Mom(LW) exhibit the largest year-to-year swings: BL-Rev(LW) reaches +46.2% in 2016 and –36.5% in 2008; BL-Mom(LW) hits +44.9% in 2024 and –11.1% in 2018. SWITCH(LW) and HRP variants show the smallest year-to-year variation in the base-strategy group, consistent with their structural design goals (diversification and regime-conditional routing, respectively).

## Appendix H — Reproducibility

This appendix documents the artifacts, software stack, and validation infrastructure required to reproduce the results in this paper from publicly available data.

### H.1 Repository structure

The paper is produced from a single git repository (next-gen-aiam) with the following top-level layout relevant to reproduction:

```

paper1/          - manuscript source (sec*.md, app*.md, glossary.md)
src/aiam/        - strategy implementations and evaluation harness
  data/          - Panel, EODHD client, returns, regimes (incl. split.py)
  estimators/    - sample_cov, ledoit_wolf_cov, oas_cov

```

```

evaluation/      - performance_stats, transaction_costs, ic.py
harness/        - run_horse_race (walk-forward dispatcher)
strategy/       - Strategy ABC + 31 base strategy classes
scripts/        - canonical build pipeline (11 scripts + generate_figures.py)
data/published/ - tracked reproduction artifacts (see below)
docs/figures/   - 11 PNGs and SVGs regenerable from generate_figures.py
notebooks/
  99_reproducibility_guard.ipynb - automated check suite

```

All strategy performance numbers in the paper are derived from the build pipeline in `scripts/` applied to EODHD daily prices. Raw prices and intermediate cache files are not tracked; the published data artifacts are the canonical source for reproduction without an EODHD API key.

---

## H.2 Published data artifacts

The `data/published/` directory contains five tracked artifacts:

| File                                       | Description                                                    |
|--------------------------------------------|----------------------------------------------------------------|
| <code>master_table_62strategies.csv</code> | Full-sample and OOS statistics for all 62 strategies           |
| <code>strategy_returns_base.parquet</code> | Daily returns for 31 base strategies, 2003-01-03 to 2026-04-30 |
| <code>strategy_returns_vmp.parquet</code>  | Daily returns for 31 VMP-overlay strategies, same period       |
| <code>ohlcv_29assets_2003_2026.csv</code>  | Daily OHLCV prices for 29 instruments (source data)            |
| <code>regime_signals.parquet</code>        | Monthly regime classifications (R0–R7) for 2003–2026           |

All paper figures and tables are reproducible from these five files. Sub-period Sharpe calculations (Appendix G, §6.3) use `strategy_returns_base.parquet` and `strategy_returns_vmp.parquet` directly; `master_table_62strategies.csv` is the canonical reference for the full-sample statistics in Appendix C.

---

## H.3 The reproducibility guard — Notebook 99

`notebooks/99_reproducibility_guard.ipynb` is an automated check suite that loads `master_table_62strategies.csv` and verifies 20 numerical claims from the paper:

- **Full-sample Sharpe spot checks** (8 checks): GMV(sample) degenerate Sharpe, EW, MDP(LW), VMP(MDP(LW)), MSR(LW), VMP(MSR(LW)), SWITCH(v2a), VMP(SWITCH(v2a)) — each checked to two decimal places against the published CSV.
- **OOS Sharpe spot checks** (3 checks): VMP(MDP(LW)) 2.422, VMP(MDP(sample)) 2.416, MDP(LW) 2.304 — test period 2023–2026.
- **VMP universal sign-test** (1 check): all 24 base–VMP pairs show positive  $\Delta$ Sharpe on both the full sample and the OOS test period; count must equal 24.

- **SWITCH(v2a) vs SWITCH(LW)** (2 checks): base Sharpe differential (+0.434) and OOS differential (+0.104).
- **Memmel test values** (2 checks): SWITCH(v2a) vs SWITCH(LW)  $z = 2.05$ ,  $p = 0.040$ ; MSR(LW) vs MSR(sample)  $z = 1.13$ ,  $p = 0.259$ .
- **Transaction cost checks** (4 checks): VMP(MDP(LW)) net 10 bps Sharpe 1.336; FF3-Mom net 10 bps Sharpe 0.394; VMP(SWITCH(LW)) net 10 bps Sharpe 1.201; FF3-Mom gross-to-net degradation 0.291.

The notebook currently passes **20/20** checks. It is the first executable to run after any change to the build pipeline or published data.

---

## H.4 The research\_agents framework

`src/aiam/research_agents/` implements the agentic governance substrate described in Paper 3. For this paper its role is methodological: it provides the deterministic validation layer that ensures reproducibility under multi-agent orchestration.

Key modules:

| Module                     | Purpose                                                                            |
|----------------------------|------------------------------------------------------------------------------------|
| <code>artifacts.py</code>  | Artifact registry — typed containers for strategy results, figures, and tables     |
| <code>manifest.py</code>   | Manifest schema — provenance tracking for all published outputs                    |
| <code>validators.py</code> | Claim-level validators — the same checks run in Notebook 99, callable from Python  |
| <code>packets.py</code>    | Agent communication schema — structured result packets for agent-to-agent handoffs |
| <code>summaries.py</code>  | Summary generators — narrative synthesis from structured result dicts              |

The framework enforces a governance-first, deterministic-core-then-orchestration pattern: numerical results are computed deterministically from the canonical pipeline; the agentic layer (summarisation, narrative generation, view synthesis in Paper 2) operates only on already-committed outputs. This separation ensures that LLM non-determinism cannot enter the quantitative claims.

---

## H.5 Software stack and random seeds

**Python version:** 3.12. **Platform:** Apple Silicon (M4 Mac, darwin); results are expected to be portable across platforms for the classical strategies (no GPU dependency). ML/DL strategies (Paper 2) require `libomp` on macOS for XGBoost and PyTorch for deep learning architectures.

**Key package versions (as of submission):**

| Package      | Version | Role                                                    |
|--------------|---------|---------------------------------------------------------|
| numpy        | 2.4.4   | Numerical core                                          |
| pandas       | 3.0.3   | Data wrangling and panel management                     |
| scipy        | 1.17.1  | Optimization (SLSQP for risk parity), statistical tests |
| scikit-learn | 1.8.0   | OAS covariance estimator, Ledoit-Wolf                   |
| cvxpy        | 1.8.2   | Convex optimization (MSR, MDP, constrained MV)          |
| xgboost      | 3.2.0   | Gradient boosting (ML track, Paper 2)                   |
| torch        | 2.12.0  | Deep learning (DL/RL tracks, Paper 2)                   |

The full dependency list is in `requirements.txt`; the installable package spec is in `pyproject.toml`.

**Random seeds.** All strategies in the 62-strategy classical comparison are deterministic: no random number generation is used. The VMP overlay, regime classifier, and all covariance-based optimizers are fully deterministic given the input price series. Sub-period Sharpe ratios and Memmel test statistics are analytically derived from the return series with no sampling step.

**Test suite.** `tests/` contains 481 pytest tests covering strategy implementations, estimators, harness logic, evaluation metrics, and the regime pipeline. All 481 tests pass on the submission codebase. Run with `pytest tests/` from the repository root with the virtual environment activated.

## H.6 Rebuilding from prices

To regenerate all results from the EODHD price data (requires `EODHD_API_KEY` environment variable):

```
python scripts/build_ohlc_29.py
python scripts/build_returns_29.py
python scripts/build_regime_signals_2003.py
python scripts/build_all_strategies_29.py
python scripts/build_weights_cache.py
python scripts/build_switch_v2a_weights.py
python scripts/generate_figures.py      # produces 11 PNGs in docs/figures/
jupyter nbconvert --to notebook --execute --inplace notebooks/99_reproducibility_guard.ipynb
```

Running `99_reproducibility_guard.ipynb` after the build confirms all 20 numerical checks pass before any paper revision is committed.

## Glossary of Acronyms

### Strategy acronyms

| Acronym  | Full name                    | Definition                                                                                                                                                              |
|----------|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BL       | Black-Litterman              | Bayesian portfolio construction framework blending an equilibrium prior with investor views.                                                                            |
| EW       | Equal Weight                 | 1/N allocation; each asset receives equal portfolio weight at each rebalance.                                                                                           |
| ERC      | Equal Risk Contribution      | Synonym for Risk Parity (RP); each asset contributes equally to total portfolio variance.                                                                               |
| FF3      | Fama-French 3-Factor         | Factor-tilt strategy family based on cross-sectional ranking by momentum, low-volatility, quality, or multi-factor composite signals; named after Fama & French (1993). |
| GMV      | Global Minimum Variance      | Portfolio that minimises total variance subject to full investment and long-only constraints.                                                                           |
| HRP      | Hierarchical Risk Parity     | Cluster-and-bisect allocation using Ward-linkage dendrogram; no matrix inversion required (de Prado 2016).                                                              |
| L/S      | Long-Short                   | Portfolio construction with both long and short legs; gross exposure = 1, net $\approx 0$ for market-neutral variants.                                                  |
| MDP      | Most Diversified Portfolio   | Maximises the Diversification Ratio $(\sigma^\top \mathbf{w}) / \sqrt{\mathbf{w}^\top \Sigma \mathbf{w}}$ (Choueifaty & Coignard 2008).                                 |
| MSR      | Maximum Sharpe Ratio         | Tangency portfolio; maximises $\hat{\mu}^\top \mathbf{w} / \sqrt{\mathbf{w}^\top \Sigma \mathbf{w}}$ over long-only weights.                                            |
| MV / MVO | Mean-Variance Optimization   | Classical Markowitz (1952) portfolio selection; balances expected return against variance.                                                                              |
| RP       | Risk Parity                  | Equal Risk Contribution; see ERC.                                                                                                                                       |
| SWITCH   | Regime-Conditional Switching | Routes to one of three base strategies (MSR(LW), MSR(sample), MDP(LW)) based on the prevailing macro regime; routing rule derived entirely from training data.          |

| Acronym | Full name                    | Definition                                                                                                                                                     |
|---------|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TSMOM   | Time-Series Momentum         | Weights assets by the sign of their trailing return scaled by target volatility (Moskowitz, Ooi & Pedersen 2012).                                              |
| VMP     | Volatility-Managed Portfolio | Daily inverse-vol overlay that scales each strategy's exposure by $\bar{\sigma}/\hat{\sigma}_t$ , clipped to $[0.25\times, 1.50\times]$ (Moreira & Muir 2017). |

### Estimator acronyms

| Acronym | Full name                      | Definition                                                                                                                                                                 |
|---------|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LW      | Ledoit-Wolf shrinkage          | Analytically optimal linear shrinkage of the sample covariance matrix toward a structured target; reduces eigenvalue noise and improves conditioning (Ledoit & Wolf 2004). |
| OAS     | Oracle Approximating Shrinkage | Non-linear shrinkage estimator minimising the Frobenius-norm distance to the oracle estimator (Chen et al. 2010).                                                          |

### Regime labels (R0–R7)

The macro classifier assigns one of eight regime labels at each month-end. Labels reflect the dominant macro character derived from GDP growth, CPI inflation, unemployment, VIX, S&P 500 trailing return, and three yield-curve features (level, slope, curvature).

| Code | Label       | Macro character                                               |
|------|-------------|---------------------------------------------------------------|
| R0   | Expansion   | Broad-based growth; equities trending; credit spreads narrow. |
| R1   | Recovery    | Post-contraction rebound; unemployment falling from peak.     |
| R2   | Neutral     | Mid-cycle; no strong directional macro signal.                |
| R3   | Slow Growth | Below-trend expansion; weak but positive GDP.                 |



| Code | Label                    | Macro character                                                                    |
|------|--------------------------|------------------------------------------------------------------------------------|
| R4   | Stress                   | Elevated VIX and/or credit spread widening without full recession.                 |
| R5   | Low Growth & Contracting | Late-cycle deceleration or early contraction; yield-curve flattening or inversion. |
| R6   | Crisis                   | Sharp drawdown in equities; spike in VIX; acute risk-off.                          |
| R7   | Contraction              | Sustained below-zero GDP growth; recession.                                        |

Regime coverage over the 2003–2026 sample (5,868 NYSE trading days): R0 27.3%; R1 25.2%; R2 3.9%; R3 1.4%; R4 3.6%; R5 15.7%; R6 2.5%; R7 20.2%. See Appendix D for the full methodology and transition matrices.

### Other acronyms

| Acronym | Full name                                                                 |
|---------|---------------------------------------------------------------------------|
| CI      | Confidence Interval                                                       |
| FRED    | Federal Reserve Economic Data (St. Louis Fed public economic data series) |
| GFC     | Global Financial Crisis (2008–2009)                                       |
| IS      | In-Sample (2003–2022 training period in this study)                       |
| NYSE    | New York Stock Exchange (trading-day calendar used throughout)            |
| OOS     | Out-of-Sample (2023–2026 test period in this study)                       |
| TC      | Transaction Costs                                                         |